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- (54) **DOSAGE AND ADMINISTRATION OF ANTI-C5 ANTIBODIES FOR TREATMENT OF PAROXYSMAL NOCTURNAL HEMOGLOBINURIA (PNH) IN PEDIATRIC PATIENTS**
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- (58) **Field of Classification Search**
None
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(57) **ABSTRACT**

Provided are methods for clinical treatment of Paroxysmal Nocturnal Hemoglobinuria (PNH) in pediatric patients using an anti-C5 antibody, or antigen binding fragment thereof.

12 Claims, No Drawings

Specification includes a Sequence Listing.

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**DOSAGE AND ADMINISTRATION OF
ANTI-C5 ANTIBODIES FOR TREATMENT
OF PAROXYSMAL NOCTURNAL
HEMOGLOBINURIA (PNH) IN PEDIATRIC
PATIENTS**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a 35 U.S.C. 371 national stage filing of International Application No. PCT/US2019/034293, filed May 29, 2019, which claims priority to, and the benefit of, U.S. Provisional Application No. 62/678,455, filed May 31, 2018. The entire contents of the aforementioned applications are incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing that has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Nov. 23, 2020, is named AXJ-245US_Sequence_Listing.txt and is 50,658 bytes in size.

BACKGROUND

Paroxysmal nocturnal hemoglobinuria (PNH) is a progressive, debilitating, and life-threatening disease characterized by complement-mediated hemolysis, thrombosis and bone marrow failure. PNH has an estimated worldwide incidence of 1.3 per million population. The onset of PNH is typically in adulthood, with pediatric cases accounting for <5% of reported cases. Given the extremely small target population, studies of children with PNH have been limited to case reports, case series, and a small clinical trial.

In adults, the clinical manifestations of PNH include hemoglobinuria, chronic renal insufficiency, erectile dysfunction, thrombosis, abdominal pain, dyspnea and dysphagia. In contrast, children with PNH usually present with nonspecific symptoms related to the underlying bone marrow disorder, such as pallor, fatigue or jaundice, with hemoglobinuria appearing less commonly. Clinical evaluation in pediatric patients also reveals bone marrow failure syndromes, such as aplastic anemia and refractory cytopenia. Once the bone marrow disorder is resolved in the child or the PNH clone expands (the cause of which is still unknown), the disease eventually evolves into one more typically seen in adults at presentation.

Thus, pediatric patients can be expected to suffer substantial morbidity related to hemolysis, as seen in adult PNH patients. Accordingly, it is an object of the present disclosure to provide improved methods for treating patients with PNH.

SUMMARY

Provided herein are compositions and methods for treating Paroxysmal Nocturnal Hemoglobinuria (PNH) in a human pediatric patient, comprising administering to the patient an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody or antigen binding fragment thereof is administered (or is for administration) according to a particular clinical dosage regimen (e.g., at a particular dose amount and according to a specific dosing schedule).

An exemplary anti-C5 antibody is ravulizumab comprising the heavy and light chains having the sequences shown

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in SEQ ID NOs:14 and 11, respectively, or antigen binding fragments and variants thereof. In other embodiments, the antibody comprises the heavy and light chain complementarity determining regions (CDRs) or variable regions (VRs) of ravulizumab. Accordingly, in one embodiment, the antibody comprises the CDR1, CDR2 and CDR3 domains of the heavy chain variable (VH) region of ravulizumab having the sequence shown in SEQ ID NO:12, and the CDR1, CDR2 and CDR3 domains of the light chain variable (VL) region of ravulizumab having the sequence shown in SEQ ID NO:8. In another embodiment, the antibody comprises CDR1, CDR2 and CDR3 heavy chain sequences as set forth in SEQ ID NOs:19, 18 and 3, respectively, and CDR1, CDR2 and CDR3 light chain sequences as set forth in SEQ ID NOs:4, 5 and 6, respectively. In another embodiment, the antibody comprises VH and VL regions having the amino acid sequences set forth in SEQ ID NO:12 and SEQ ID NO:8, respectively. In another embodiment, the antibody comprises a heavy chain constant region as set forth in SEQ ID NO:13.

In another embodiment, the antibody comprises a variant human Fc constant region that binds to human neonatal Fc receptor (FcRn), wherein the variant human Fc CH3 constant region comprises Met429Leu and Asn435Ser substitutions at residues corresponding to methionine 428 and asparagine 434 of a native human IgG Fc constant region, each according to the EU numbering convention.

In another embodiment, the antibody comprises CDR1, CDR2 and CDR3 heavy chain sequences as set forth in SEQ ID NOs:19, 18 and 3, respectively, and CDR1, CDR2 and CDR3 light chain sequences as set forth in SEQ ID NOs:4, 5 and 6, respectively and a variant human Fc constant region that binds to human neonatal Fc receptor (FcRn), wherein the variant human Fc CH3 constant region comprises Met429Leu and Asn435Ser substitutions at residues corresponding to methionine 428 and asparagine 434 of a native human IgG Fc constant region, each according to the EU numbering convention.

In another embodiment, the anti-C5 antibody comprises the heavy and light chain CDRs or variable regions of the BNJ421 antibody (described in PCT/US2015/019225 and U.S. Pat. No. 9,079,949). In another embodiment, the anti-C5 antibody comprises the heavy and light chain CDRs or variable regions of the 7086 antibody (see U.S. Pat. Nos. 8,241,628 and 8,883,158). In another embodiment, the anti-C5 antibody comprises the heavy and light chain CDRs or variable regions of the 8110 antibody (see U.S. Pat. Nos. 8,241,628 and 8,883,158). In another embodiment, the anti-C5 antibody comprises the heavy and light chain CDRs or variable regions of the 305LO5 antibody (see US2016/0176954A1). In another embodiment, the anti-C5 antibody comprises the heavy and light chain CDRs or variable regions of the SKY59 antibody.

In another embodiment, the antibody competes for binding with, and/or binds to the same epitope on C5 as any of the above-mentioned antibodies. In another embodiment, the antibody has at least about 90% variable region amino acid sequence identity to any of the above-mentioned antibodies (e.g., at least about 90%, 95% or 99% variable region identity with SEQ ID NO:12 or SEQ ID NO:8).

In another embodiment, the antibody binds to human C5 at pH 7.4 and 25 C with an affinity dissociation constant (K_D) that is in the range $0.1 \text{ nM} \leq K_D \leq 1 \text{ nM}$. In another embodiment, the antibody binds to human C5 at pH 6.0 and 25 C with a $K_D \geq 10 \text{ nM}$. In yet another embodiment, the $[(K_D \text{ of the antibody or antigen-binding fragment thereof for human C5 at pH 6.0 and at 25 C}) / (K_D \text{ of the antibody or$

antigen-binding fragment thereof for human C5 at pH 7.4 and at 25 C)] of the antibody is greater than 25.

In one embodiment, the dose of the anti-C5 antibody, or antigen binding fragment thereof, is based on the weight of the patient. In one embodiment, for example, 300 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 5 to <10 kg. In another embodiment, 600 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 10 to <20 kg. In another embodiment, 900 mg or 2100 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 20 to <30 kg. In another embodiment, 1200 mg or 2700 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 30 to <40 kg. In another embodiment, 2400 mg or 3000 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 40 to <60 kg. In another embodiment, 2700 mg or 3300 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 60 to <100 kg. In another embodiment, 3000 mg or 3600 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 100 kg. In certain embodiments, dosage regimens are adjusted to provide the optimum desired response (e.g., an effective response).

In another embodiment, the anti-C5 antibody or antigen binding fragment thereof is administered for one or more administration cycles. In one embodiment, the treatment (e.g., administration cycle) is 26 weeks. In one embodiment, the anti-C5 antibody or antigen binding fragment thereof is administered once on Day 1 (e.g., of the administration cycle), once on Day 15 (e.g., of the administration cycle), and every four weeks thereafter. In another embodiment, the anti-C5 antibody or antigen binding fragment thereof is administered once on Day 1 (e.g., of the administration cycle), once on Day 15 (e.g., of the administration cycle), and every eight weeks thereafter. In another embodiment, the anti-C5 antibody or antigen binding fragment thereof is administered every four or eight weeks after treatment (e.g., an administration cycle) for an extension period up to two years (e.g., at a dose of 300 mg, 600 mg, 900 mg, 1200 mg, 2100 mg, 2400 mg, 2700 mg, 3000 mg, 3300 mg or 3600 mg).

In another embodiment, a method of treating a human patient with PNH is provided, the method comprising administering to the patient (e.g., during an administration cycle) an effective amount of an anti-C5 antibody or antigen binding fragment thereof, comprising CDR1, CDR2 and CDR3 heavy chain sequences as set forth in SEQ ID NOs:19, 18 and 3, respectively, and CDR1, CDR2 and CDR3 light chain sequences as set forth in SEQ ID NOs:4, 5 and 6, respectively, wherein the anti-C5 antibody or antigen binding fragment thereof is administered: (a) once on Day 1 at a dose of 600 mg to a patient weighing ≥ 5 to <10 kg, 600 mg to a patient weighing ≥ 10 to <20 kg, 900 mg to a patient weighing ≥ 20 to <30 kg, 1200 mg to a patient weighing ≥ 30 to <40 kg, 2400 mg to a patient weighing ≥ 40 to <60 kg, 2700 mg to a patient weighing ≥ 60 to <100 kg, or 3000 mg to a patient weighing ≥ 100 kg; and (b) on Day 15 and every four weeks thereafter at a dose of 300 mg to a patient weighing ≥ 5 to <10 kg or 600 mg to a patient weighing ≥ 10 to <20 kg; or on Day 15 and every eight weeks thereafter at a dose of 2100 mg to a patient weighing ≥ 20 to <30 kg, 2700 mg to a patient weighing ≥ 30 to <40 kg, 3000 mg to a patient weighing ≥ 40 to <60 kg, 3300 mg to a patient weighing ≥ 60 to <100 kg, or 3600 mg to a patient weighing ≥ 100 kg.

In another embodiment, a method of treating a human patient with PNH is provided, the method comprising administering to the patient (e.g., during an administration cycle) an effective amount of an anti-C5 antibody or antigen binding fragment thereof comprising CDR1, CDR2 and CDR3 heavy chain sequences as set forth in SEQ ID NOs:19, 18 and 3, respectively, CDR1, CDR2 and CDR3 light chain sequences as set forth in SEQ ID NOs:4, 5 and 6, respectively, and a variant human Fc constant region that binds to human neonatal Fc receptor (FcRn), wherein the variant human Fc CH3 constant region comprises Met429Leu and Asn435Ser substitutions at residues corresponding to methionine 428 and asparagine 434 of a native human IgG Fc constant region, each according to the EU numbering convention, wherein the anti-C5 antibody or antigen binding fragment thereof is administered: (a) once on Day 1 at a dose of 600 mg to a patient weighing ≥ 5 to <10 kg, 600 mg to a patient weighing ≥ 10 to <20 kg, 900 mg to a patient weighing ≥ 20 to <30 kg, 1200 mg to a patient weighing ≥ 30 to <40 kg, 2400 mg to a patient weighing ≥ 40 to <60 kg, 2700 mg to a patient weighing ≥ 60 to <100 kg, or 3000 mg to a patient weighing ≥ 100 kg; and (b) on Day 15 and every four weeks thereafter at a dose of 300 mg to a patient weighing ≥ 5 to <10 kg or 600 mg to a patient weighing ≥ 10 to <20 kg; or on Day 15 and every eight weeks thereafter at a dose of 2100 mg to a patient weighing ≥ 20 to <30 kg, 2700 mg to a patient weighing ≥ 30 to <40 kg, 3000 mg to a patient weighing ≥ 40 to <60 kg, 3300 mg to a patient weighing ≥ 60 to <100 kg, or 3600 mg to a patient weighing ≥ 100 kg.

In another embodiment, the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 5 to <10 kg: (a) once on Day 1 at a dose of 600 mg; and (b) on Day 15 and every four weeks thereafter at a dose of 300 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 10 to <20 kg: (a) once on Day 1 at a dose of 600 mg; and (b) on Day 15 and every four weeks thereafter at a dose of 600 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 20 to <30 kg: (a) once on Day 1 at a dose of 900 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 2100 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 30 to <40 kg: (a) once on Day 1 at a dose of 1200 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 2700 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 40 to <60 kg: (a) once on Day 1 at a dose of 2400 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3000 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 60 to <100 kg: (a) once on Day 1 at a dose of 2700 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3300 mg.

In another embodiment, the anti-C5 antibody is administered to a patient weighing ≥ 100 kg: (a) once on Day 1 at a dose of 3000 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3600 mg.

In one embodiment, the patient has not previously been treated with eculizumab. In another embodiment, the patient has previously been treated with eculizumab. In another embodiment, the patient has previously been treated with eculizumab and Day 1 (e.g., of the administration cycle) is two weeks or more from the patient's last dose of eculizumab.

In another aspect, the treatment regimens described are sufficient to maintain particular serum trough concentrations of the anti-C5 antibody or antigen binding fragment thereof. In one embodiment, for example, the treatment regimen maintains a serum trough concentration of the anti-C5 antibody or antigen binding fragment thereof of 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 200, 205, 210, 215, 220, 225, 230, 240, 245, 250, 255, 260, 265, 270, 280, 290, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395 or 400 $\mu\text{g/mL}$ or greater. In one embodiment, the treatment regimen maintains a serum trough concentration of the anti-C5 antibody or antigen binding fragment thereof of 100 $\mu\text{g/mL}$ or greater, 150 $\mu\text{g/mL}$ or greater, 200 $\mu\text{g/mL}$ or greater, 250 $\mu\text{g/mL}$ or greater, or 300 $\mu\text{g/mL}$ or greater. In another embodiment, the treatment maintains a serum trough concentration of the anti-C5 antibody or antigen binding fragment thereof of between 100 $\mu\text{g/mL}$ and 200 $\mu\text{g/mL}$. In another embodiment, the treatment maintains a serum trough concentration of the anti-C5 antibody or antigen binding fragment thereof of about 175 $\mu\text{g/mL}$.

In another embodiment, to obtain an effective response, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain at least 50 μg , 55 μg , 60 μg , 65 μg , 70 μg , 75 μg , 80 μg , 85 μg , 90 μg , 95 μg , 100 μg , 105 μg , 110 μg , 115 μg , 120 μg , 125 μg , 130 μg , 135 μg , 140 μg , 145 μg , 150 μg , 155 μg , 160 μg , 165 μg , 170 μg , 175 μg , 180 μg , 185 μg , 190 μg , 195 μg , 200 μg , 205 μg , 210 μg , 215 μg , 220 μg , 225 μg , 230 μg , 235 μg , 240 μg , 245 μg , 250 μg , 255 μg or 260 μg of antibody per milliliter of the patient's blood. In another embodiment, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain between 50 μg and 250 μg of antibody per milliliter of the patient's blood. In another embodiment, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain between 100 μg and 200 μg of antibody per milliliter of the patient's blood. In another embodiment, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain about 175 μg of antibody per milliliter of the patient's blood.

In another embodiment, to obtain an effective response, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain a minimum free C5 concentration. In one embodiment, for example, the anti-C5 antibody is administered to the patient in an amount and with a frequency to maintain a free C5 concentration of 0.2 $\mu\text{g/mL}$, 0.3 $\mu\text{g/mL}$, 0.4 $\mu\text{g/mL}$, 0.5 $\mu\text{g/mL}$ or less.

The anti-C5 antibodies, or antigen binding fragments thereof, can be administered to a patient by any suitable means. In one embodiment, the antibodies are formulated for intravenous administration.

The efficacy of the treatment methods provided herein can be assessed using any suitable means. In one embodiment, for a pediatric PNH patient, the treatment produces at least one therapeutic effect selected from the group consisting of: a reduction or cessation in fatigue, abdominal pain, dyspnea, dysphagia and chest pain compared to baseline. In another embodiment, the treatment results in terminal complement inhibition. In another embodiment, the treatment results in a reduction of hemolysis as assessed by lactate dehydrogenase (LDH) levels compared to baseline. In another embodiment, the treatment produces a shift toward normal levels of at least one hemolysis-related hematologic biomarker selected, for example, from the group consisting of: free hemoglobin, haptoglobin, reticulocyte count, PNH red blood cell (RBC)

clone and D-dimer. In another embodiment, the treatment produces a reduction in the need for blood transfusions compared to baseline. In another embodiment, the treatment produces a reduction in major adverse vascular events (MAVEs). In another embodiment, the treatment produces a shift toward normal levels of a chronic disease associated biomarker selected from the group consisting estimated glomerular filtration rate (eGFR) and spot urine:albumin:creatinine and plasma brain natriuretic peptide (BNP). In another embodiment, the treatment produces a change from baseline in quality of life as assessed via the Functional Assessment of Chronic Illness Therapy (FACIT)-Fatigue Scale, version 4 and the European Organisation for Research and Treatment of Cancer, Quality of Life Questionnaire-Core 30 Scale compared to baseline.

In a particular embodiment, LDH levels are used to evaluate responsiveness to a therapy (e.g., a reduction of hemolysis as assessed by LDH levels is indicative of an improvement in at least one sign of PNH). In one embodiment, patients treated according to the disclosed methods experience reductions in LDH levels to near normal levels or to within 10%, or within 20% above what is considered the normal level (e.g., within 105-333 IU/L (international units per liter)). In another embodiment, the patient's LDH levels are normalized throughout maintenance period of treatment. In another embodiment, the treated patient's LDH levels are normalized at least at least 95% of the time while on the maintenance period of treatment. In another embodiment, the treated patient's LDH levels are normalized at least at least 90%, 85% or 80% of the time while on the maintenance period of treatment. In one embodiment, the patient's LDH levels are ≥ 1.5 fold above the upper limit of normal ($\text{LDH} \geq 1.5 \times \text{ULN}$) prior to initiating treatment.

In one embodiment, patients treated according to the disclosed methods experience reductions in LDH levels to within normal levels or to within 10%, 20%, 30%, 40% or within 50% below what is considered the ULN level (e.g., within 105-333 IU/L (international units per liter)). In one embodiment, the patient's LDH levels are ≥ 1.5 fold above the ULN ($\text{LDH} \geq 1.5 \times \text{ULN}$) prior to initiating treatment.

In another aspect, an anti-C5 antibody or antigen binding fragment thereof is provided, comprising CDR1, CDR2 and CDR3 domains of the heavy chain variable region having the sequence set forth in SEQ ID NO:12, and CDR1, CDR2 and CDR3 domains of the light chain variable region having the sequence set forth in SEQ ID NO:8, for administration: (a) once on Day 1 at a dose of 600 mg to a patient weighing ≥ 5 to <10 kg, 600 mg to a patient weighing ≥ 10 to <20 kg, 900 mg to a patient weighing ≥ 20 to <30 kg, 1200 mg to a patient weighing ≥ 30 to <40 kg, 2400 mg to a patient weighing ≥ 40 to <60 kg, 2700 mg to a patient weighing ≥ 60 to <100 kg, or 3000 mg to a patient weighing ≥ 100 kg; and (b) on Day 15 and every four weeks thereafter at a dose of 300 mg to a patient weighing ≥ 5 to <10 kg or 600 mg to a patient weighing ≥ 10 to <20 kg; or on Day 15 and every eight weeks thereafter at a dose of 2100 mg to a patient weighing ≥ 20 to <30 kg, 2700 mg to a patient weighing 30 to <40 kg, 3000 mg to a patient weighing 40 to <60 kg, 3300 mg to a patient weighing 60 to <100 kg, or 3600 mg to a patient weighing 100 kg.

In one embodiment, the antibody is determined to be safe, tolerable and sufficiently non-immunogenic after multiple IV doses for use in PNH pediatric patients.

Further provided are kits that include a pharmaceutical composition containing an anti-C5 antibody or antigen binding fragment thereof, such as ravulizumab, and a pharmaceutically acceptable carrier, in a therapeutically effective

amount adapted for use in the methods described herein. In one embodiment, the kit comprises: (a) a dose of an anti-C5 antibody or antigen binding fragment thereof, comprising CDR1, CDR2 and CDR3 domains of the heavy chain variable region having the sequence set forth in SEQ ID NO:12, and CDR1, CDR2 and CDR3 domains of the light chain variable region having the sequence set forth in SEQ ID NO:8; and (b) instructions for using the anti-C5 antibody or antigen binding fragment thereof in the methods described herein.

In one embodiment, 300 mg or 600 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 5 to <10 kg. In another embodiment, 600 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 10 to <20 kg. In another embodiment, 900 mg or 2100 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 20 to <30 kg. In another embodiment, 1200 mg or 2700 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 30 to <40 kg. In another embodiment, 2400 mg or 3000 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 40 to <60 kg. In another embodiment, 2700 mg or 3300 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 60 to <100 kg. In another embodiment, 3000 mg or 3600 mg of the anti-C5 antibody or antigen binding fragment thereof is administered to a patient weighing ≥ 100 kg.

DETAILED DESCRIPTION

I. Definitions

As used herein, the term “subject” or “patient” is a human patient (e.g., a patient having Paroxysmal Nocturnal Hemoglobinuria (PNH)).

As used herein, the term “pediatric” patient is a human patient under 18 years of age (<18 years of age).

PNH is an acquired hemolytic disorder that occurs most frequently in adults (Brodsky, R., *Blood*, 126:2459-65, 2015). The disease begins with the clonal expansion of a hematopoietic stem cell that has acquired a somatic mutation in the PIGA gene (Brodsky, R., *Blood*, 124:2804-11, 2014). Consequently, PNH blood cells lack the glycosphosphatidylinositol (GPI) anchor protein and are deficient in the membrane-bound complement inhibitory proteins CD55 and CD59. In the absence of CD55, there is increased deposition of complement protein C3 cleavage products on blood cell membrane surfaces, in turn leading to cleavage of C5 into C5a and C5b. The pathology and clinical presentations in patients with PNH are driven by uncontrolled terminal complement activation.

C5a is a potent anaphylatoxin, chemotactic factor, and cell-activating molecule that mediates multiple pro-inflammatory and pro-thrombotic activities (Matis, L & Rollins, S., *Nat. Med.*, 1:839-42, 1995; Prodinger et al., Complement. In: Paul W E, editor. *Fundamental immunology* (4th ed). Philadelphia: Lippincott-Raven Publishers; 1999. p. 967-95). C5b recruits the terminal complement components C6, C7, C8 and C9 to form the pro-inflammatory, pro-thrombotic cytolytic pore molecule C5b-9, a process that under normal circumstances would be blocked on the red blood cell (RBC) membrane by CD59. In patients with PNH, however, these final steps proceed unchecked, culminating in hemolysis and the release of free hemoglobin, as well as platelet activation (Hill, A. et al., *Blood*, 121:4985-96,

2013). The signs and symptoms of PNH can be attributed to chronic, uncontrolled complement C5 cleavage, and release of C5a and C5b-9 leading to RBC hemolysis, which together result in:

- 5 release of intracellular free hemoglobin and lactate dehydrogenase (LDH) into circulation as a direct consequence of hemolysis;
- irreversible binding to and inactivation of nitric oxide (NO) by hemoglobin, and inhibition of NO synthesis;
- 10 vasoconstriction and tissue-bed ischemia due to absence of vasodilatory NO, as well as possible microthrombi manifesting as abdominal pain, dysphagia and erectile dysfunction;
- platelet activation; and
- 15 a pro-inflammatory and prothrombotic state.

A substantial proportion of patients with PNH experience renal dysfunction and pulmonary hypertension (Hillmen, P. et al., *Am. J. Hematol.*, 85:553-9, 2010 [erratum in *Am. J. Hematol.*, 85:911, 2010]; Hill, A. et al., *Br. J. Haematol.*, 158:409-14, 2012). Patients also experience venous or arterial thrombosis in diverse sites, including the abdomen or central nervous system.

In contrast, children with PNH usually present with nonspecific symptoms related to the underlying bone marrow disorder, such as pallor, fatigue or jaundice, with hemoglobinuria appearing less commonly (Ware, R. et al., *N. Engl. J. Med.*, 325:991-6, 1991). Clinical evaluation in pediatric patients also reveals bone marrow failure syndromes, such as, for example, aplastic anemia and refractory cytopenia (van den Heuvel-Eibrink, M., *Paediatr. Drugs*, 9:11-6, 2007). Once the bone marrow disorder is resolved in the child or the PNH clone expands (the cause of which is still unknown), the disease eventually evolves into one more typically seen in adults at presentation.

As used herein, “effective treatment” refers to treatment producing a beneficial effect, e.g., amelioration of at least one symptom of a disease or disorder. A beneficial effect can take the form of an improvement over baseline, e.g., an improvement over a measurement or observation made prior to initiation of therapy according to the method. Effective treatment may refer to alleviation of at least one symptom of PNH (e.g., pallor, fatigue, jaundice, anemia, cytopenia, abdominal pain, dyspnea, dysphagia, chest pain or erectile dysfunction).

45 The term “effective amount” refers to an amount of an agent that provides the desired biological, therapeutic and/or prophylactic result. That result can be reduction, amelioration, palliation, lessening, delaying and/or alleviation of one or more of the signs, symptoms or causes of a disease, or any other desired alteration of a biological system. In one example, an “effective amount” is the amount of anti-C5 antibody, or antigen binding fragment thereof, clinically proven to alleviate at least one symptom of PNH (e.g., pallor, fatigue, jaundice, anemia, cytopenia, abdominal pain, dyspnea, dysphagia, or chest pain). An effective amount can be administered in one or more administrations.

As used herein, the term “loading dose” refers to the first dose administered (e.g., during an administration cycle).

As used herein, the terms “maintenance” and “maintenance phase” are used interchangeably and refer to the second phase of treatment. In certain embodiments, treatment is continued as long as clinical benefit is observed or until unmanageable toxicity or disease progression occurs.

As used herein, the term “serum trough level” refers to the lowest level that the agent (e.g., the anti-C5 antibody, or antigen binding fragment thereof) or medicine is present in the serum. In contrast, a “peak serum level,” refers to the

highest level of the agent in the serum. The “average serum level,” refers to the mean level of the agent in the serum over time.

The term “antibody” describes a polypeptide comprising at least one antibody-derived antigen binding site (e.g., VH/VL region or Fv, or CDR). Antibodies include known forms of antibodies, e.g., the antibody can be a human antibody, a humanized antibody, a bispecific antibody or a chimeric antibody. The antibody also can be a Fab, Fab’2, ScFv, SMIP, Affibody®, nanobody or a single-domain antibody. The antibody also can be of any of the following isotypes: IgG1, IgG2, IgG3, IgG4, IgM, IgA1, IgA2, IgAsec, IgD, IgE or combinations thereof. The antibody can be a naturally occurring antibody or an antibody that has been altered by a protein engineering technique (e.g., by mutation, deletion, substitution, conjugation to a non-antibody moiety). An antibody can include, for example, one or more variant amino acids (compared to a naturally occurring antibody) that change a property (e.g., a functional property) of the antibody. Numerous such alterations are known in the art that affect, e.g., half-life, effector function, and/or immune responses to the antibody in a patient. The term antibody also includes artificial or engineered polypeptide constructs that comprise at least one antibody-derived antigen binding site.

II. Anti-C5 Antibodies

Anti-C5 antibodies described herein bind to complement component C5 (e.g., human C5) and inhibit the cleavage of C5 into fragments C5a and C5b. As described above, such antibodies also have, for example, improved pharmacokinetic properties relative to other anti-C5 antibodies (e.g., eculizumab) used for therapeutic purposes.

Anti-C5 antibodies (or VH/VL domains derived therefrom) suitable for use in the methods described herein can be generated using methods known in the art. Alternatively, art recognized anti-C5 antibodies can be used. Antibodies that compete for binding to C5 with any of these art recognized antibodies or antibodies described herein can also be used.

An exemplary anti-C5 antibody is ravulizumab comprising heavy and light chains having the sequences shown in SEQ ID NOs:14 and 11, respectively, or antigen binding fragments and variants thereof. Ravulizumab (also known as BNJ441 and ALXN1210) is described in PCT/US2015/019225 and U.S. Pat. No. 9,079,949, the entire teachings of which are hereby incorporated by reference. The terms ravulizumab, BNJ441, and ALXN1210 may be used interchangeably throughout this document, but all refer to the same antibody. Ravulizumab selectively binds to human complement protein C5, inhibiting its cleavage to C5a and C5b during complement activation. This inhibition prevents the release of the proinflammatory mediator C5a and the formation of the cytolytic pore-forming membrane attack complex (MAC) C5b-9 while preserving the proximal or early components of complement activation (e.g., C3 and C3b) essential for the opsonization of microorganisms and clearance of immune complexes.

In other embodiments, the antibody comprises the heavy and light chain CDRs or variable regions of ravulizumab. Accordingly, in one embodiment, the antibody comprises the CDR1, CDR2 and CDR3 domains of the VH region of ravulizumab having the sequence set forth in SEQ ID NO:12, and the CDR1, CDR2 and CDR3 domains of the VL region of ravulizumab having the sequence set forth in SEQ ID NO:8. In another embodiment, the antibody comprises heavy chain CDR1, CDR2 and CDR3 domains having the

sequences set forth in SEQ ID NOs:19, 18 and 3, respectively, and light chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:4, 5 and 6, respectively. In another embodiment, the antibody comprises VH and VL regions having the amino acid sequences set forth in SEQ ID NO:12 and SEQ ID NO:8, respectively.

Another exemplary anti-C5 antibody is antibody BNJ421 comprising heavy and light chains having the sequences shown in SEQ ID NOs:20 and 11, respectively, or antigen binding fragments and variants thereof. BNJ421 (also known as ALXN1211) is described in PCT/US2015/019225 and U.S. Pat. No. 9,079,949, the entire teachings of which are hereby incorporated by reference.

In other embodiments, the antibody comprises the heavy and light chain CDRs or variable regions of BNJ421. Accordingly, in one embodiment, the antibody comprises the CDR1, CDR2 and CDR3 domains of the VH region of BNJ421 having the sequence set forth in SEQ ID NO:12, and the CDR1, CDR2 and CDR3 domains of the VL region of BNJ421 having the sequence set forth in SEQ ID NO:8. In another embodiment, the antibody comprises heavy chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:19, 18 and 3, respectively, and light chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:4, 5 and 6, respectively. In another embodiment, the antibody comprises VH and VL regions having the amino acid sequences set forth in SEQ ID NO:12 and SEQ ID NO:8, respectively.

The exact boundaries of CDRs are defined differently according to different methods. In some embodiments, the positions of the CDRs or framework regions within a light or heavy chain variable domain are as defined by Kabat et al. [(1991) “Sequences of Proteins of Immunological Interest.” NIH Publication No. 91-3242, U.S. Department of Health and Human Services, Bethesda, MD]. In such cases, the CDRs can be referred to as “Kabat CDRs” (e.g., “Kabat LCDR2” or “Kabat HCDR1”). In some embodiments, the positions of the CDRs of a light or heavy chain variable region are as defined by Chothia et al. (*Nature*, 342:877-83, 1989). Accordingly, these regions can be referred to as “Chothia CDRs” (e.g., “Chothia LCDR2” or “Chothia HCDR3”). In some embodiments, the positions of the CDRs of the light and heavy chain variable regions can be defined by a Kabat-Chothia combined definition. In such embodiments, these regions can be referred to as “combined Kabat-Chothia CDRs.” Thomas, C. et al. (*Mol. Immunol.*, 33:1389-401, 1996) exemplifies the identification of CDR boundaries according to Kabat and Chothia numbering schemes.

Another exemplary anti-C5 antibody is the 7086 antibody described in U.S. Pat. Nos. 8,241,628 and 8,883,158. In one embodiment, the antibody comprises the heavy and light chain CDRs or variable regions of the 7086 antibody (see U.S. Pat. Nos. 8,241,628 and 8,883,158). In another embodiment, the antibody, or antigen binding fragment thereof, comprises heavy chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:21, 22 and 23, respectively, and light chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:24, 25 and 26, respectively. In another embodiment, the antibody, or antigen binding fragment thereof, comprises the VH region of the 7086 antibody having the sequence set forth in SEQ ID NO:27, and the VL region of the 7086 antibody having the sequence set forth in SEQ ID NO:28.

Another exemplary anti-C5 antibody is the 8110 antibody also described in U.S. Pat. Nos. 8,241,628 and 8,883,158. In one embodiment, the antibody comprises the heavy and light chain CDRs or variable regions of the 8110 antibody. In

another embodiment, the antibody, or antigen binding fragment thereof, comprises heavy chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:29, 30 and 31, respectively, and light chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:32, 33 and 34, respectively. In another embodiment, the antibody comprises the VH region of the 8110 antibody having the sequence set forth in SEQ ID NO:35, and the VL region of the 8110 antibody having the sequence set forth in SEQ ID NO:36.

Another exemplary anti-C5 antibody is the 305LO5 antibody described in US2016/0176954A1. In one embodiment, the antibody comprises the heavy and light chain CDRs or variable regions of the 305LO5 antibody. In another embodiment, the antibody, or antigen binding fragment thereof, comprises heavy chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:37, 38 and 39, respectively, and light chain CDR1, CDR2 and CDR3 domains having the sequences set forth in SEQ ID NOs:40, 41 and 42, respectively. In another embodiment, the antibody comprises the VH region of the 305LO5 antibody having the sequence set forth in SEQ ID NO:43, and the VL region of the 305LO5 antibody having the sequence set forth in SEQ ID NO:44.

Another exemplary anti-C5 antibody is the SKY59 antibody (Fukuzawa, T. et al., *Sci. Rep.*, 7:1080, 2017). In one embodiment, the antibody comprises the heavy and light chain CDRs or variable regions of the SKY59 antibody. In another embodiment, the antibody, or antigen binding fragment thereof, comprises a heavy chain comprising SEQ ID NO:45 and a light chain comprising SEQ ID NO:46.

In some embodiments, an anti-C5 antibody described herein comprises a heavy chain CDR1 comprising, or consisting of, the following amino acid sequence: GHIFSNIWIQ (SEQ ID NO:19). In some embodiments, an anti-C5 antibody described herein comprises a heavy chain CDR2 comprising, or consisting of, the following amino acid sequence: EILPGSGHTEYTENFKD (SEQ ID NO:18). In some embodiments, an anti-C5 antibody described herein comprises a heavy chain variable region comprising the following amino acid sequence:

(SEQ ID NO: 12)

QVQLVQSGAE VKKPGASVKV SCKASGHIFS NYWIQWVRQA
 PGQGLEWMGE ILPGSGHTEY TENFKDRVIM TRDTSTSTVY
 MELSSLRSED TAVYYCARYF FGSSPNWYFD VWGQGTLTVT
 SS.

In some embodiments, an anti-C5 antibody described herein comprises a light chain variable region comprising the following amino acid sequence:

(SEQ ID NO: 8)

DIQMTQSPSS LSASVGRVIT ITCGASENIY GALNWYQQKP
 GKAPKLLIYG ATNLADGVPS RFGSGSGSTD FTLTISSLQP
 EDFATYYCQN VLNTPLIFGQ GTKVEIK.

An anti-C5 antibody described herein can, in some embodiments, comprise a variant human Fc constant region that binds to human neonatal Fc receptor (FcRn) with greater affinity than that of the native human Fc constant region from which the variant human Fc constant region was derived. The Fc constant region can, for example, comprise

one or more (e.g., two, three, four, five, six, seven, or eight or more) amino acid substitutions relative to the native human Fc constant region from which the variant human Fc constant region was derived. The substitutions can increase the binding affinity of an IgG antibody containing the variant Fc constant region to FcRn at pH 6.0, while maintaining the pH dependence of the interaction. Methods for testing whether one or more substitutions in the Fc constant region of an antibody increase the affinity of the Fc constant region for FcRn at pH 6.0 (while maintaining pH dependence of the interaction) are known in the art and exemplified in the working examples. See, e.g., PCT/US2015/019225 and U.S. Pat. No. 9,079,949 the disclosures of each of which are incorporated herein by reference in their entirety.

Substitutions that enhance the binding affinity of an antibody Fc constant region for FcRn are known in the art and include, e.g., (1) the M252Y/S254T/T256E triple substitution (Dall'Acqua, W. et al., *J. Biol. Chem.*, 281:23514-24, 2006); (2) the M428L or T250Q/M428L substitutions (Hinton, P. et al., *J. Biol. Chem.*, 279:6213-6, 2004; Hinton, P. et al., *J. Immunol.*, 176:346-56, 2006); and (3) the N434A or T307/E380A/N434A substitutions (Petkova, S. et al., *Int. Immunol.*, 18:1759-69, 2006). The additional substitution pairings: P257I/Q311I, P257I/N434H and D376V/N434H (Datta-Mannan, A. et al., *J. Biol. Chem.*, 282:1709-17, 2007), the disclosures of each of which are incorporated herein by reference in their entirety.

In some embodiments, the variant constant region has a substitution at EU amino acid position 255 for valine. In some embodiments, the variant constant region has a substitution at EU amino acid position 309 for asparagine. In some embodiments, the variant constant region has a substitution at EU amino acid position 312 for isoleucine. In some embodiments, the variant constant region has a substitution at EU amino acid position 386.

In some embodiments, the variant Fc constant region comprises no more than 30 (e.g., no more than 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3 or 2) amino acid substitutions, insertions, or deletions relative to the native constant region from which it was derived. In some embodiments, the variant Fc constant region comprises one or more amino acid substitutions selected from the group consisting of: M252Y, S254T, T256E, N434S, M428L, V259I, T250I and V308F. In some embodiments, the variant human Fc constant region comprises a methionine at position 428 and an asparagine at position 434 of a native human IgG Fc constant region, each in EU numbering. In some embodiments, the variant Fc constant region comprises a 428L/434S double substitution as described in, e.g., U.S. Pat. No. 8,088,376.

In some embodiments the precise location of these mutations may be shifted from the native human Fc constant region position due to antibody engineering. For example, the 428L/434S double substitution when used in a IgG2/4 chimeric Fc may correspond to 429L and 435S as in the M429L and N435S variants found in ravulizumab and described in U.S. Pat. No. 9,079,949 the disclosure of which is incorporated herein by reference in its entirety.

In some embodiments, the variant constant region comprises a substitution at amino acid position 237, 238, 239, 248, 250, 252, 254, 255, 256, 257, 258, 265, 270, 286, 289, 297, 298, 303, 305, 307, 308, 309, 311, 312, 314, 315, 317, 325, 332, 334, 360, 376, 380, 382, 384, 385, 386, 387, 389, 424, 428, 433, 434 or 436 (EU numbering) relative to the native human Fc constant region. In some embodiments, the substitution is selected from the group consisting of: methio-

nine for glycine at position 237; alanine for proline at position 238; lysine for serine at position 239; isoleucine for lysine at position 248; alanine, phenylalanine, isoleucine, methionine, glutamine, serine, valine, tryptophan, or tyrosine for threonine at position 250; phenylalanine, tryptophan, or tyrosine for methionine at position 252; threonine for serine at position 254; glutamic acid for arginine at position 255; aspartic acid, glutamic acid, or glutamine for threonine at position 256; alanine, glycine, isoleucine, leucine, methionine, asparagine, serine, threonine, or valine for proline at position 257; histidine for glutamic acid at position 258; alanine for aspartic acid at position 265; phenylalanine for aspartic acid at position 270; alanine, or glutamic acid for asparagine at position 286; histidine for threonine at position 289; alanine for asparagine at position 297; glycine for serine at position 298; alanine for valine at position 303; alanine for valine at position 305; alanine, aspartic acid, phenylalanine, glycine, histidine, isoleucine, lysine, leucine, methionine, asparagine, proline, glutamine, arginine, serine, valine, tryptophan, or tyrosine for threonine at position 307; alanine, phenylalanine, isoleucine, leucine, methionine, proline, glutamine, or threonine for valine at position 308; alanine, aspartic acid, glutamic acid, proline, or arginine for leucine or valine at position 309; alanine, histidine, or isoleucine for glutamine at position 311; alanine or histidine for aspartic acid at position 312; lysine or arginine for leucine at position 314; alanine or histidine for asparagine at position 315; alanine for lysine at position 317; glycine for asparagine at position 325; valine for isoleucine at position 332; leucine for lysine at position 334; histidine for lysine at position 360; alanine for aspartic acid at position 376; alanine for glutamic acid at position 380; alanine for glutamic acid at position 382; alanine for asparagine or serine at position 384; aspartic acid or histidine for glycine at position 385; proline for glutamine at position 386; glutamic acid for proline at position 387; alanine or serine for asparagine at position 389; alanine for serine at position 424; alanine, aspartic acid, phenylalanine, glycine, histidine, isoleucine, lysine, leucine, asparagine, proline, glutamine, serine, threonine, valine, tryptophan, or tyrosine for methionine at position 428; lysine for histidine at position 433; alanine, phenylalanine, histidine, serine, tryptophan, or tyrosine for asparagine at position 434; and histidine for tyrosine or phenylalanine at position 436, all in EU numbering.

Suitable anti-C5 antibodies for use in the methods described herein, in some embodiments, comprise a heavy chain polypeptide comprising the amino acid sequence set forth in SEQ ID NO:14 and/or a light chain polypeptide comprising the amino acid sequence set forth in SEQ ID NO:11. Alternatively, the anti-C5 antibodies for use in the methods described herein, in some embodiments, comprise a heavy chain polypeptide comprising the amino acid sequence set forth in SEQ ID NO:20 and/or a light chain polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 11.

In one embodiment, the antibody binds to C5 at pH 7.4 and 25 C (and, otherwise, under physiologic conditions) with an affinity dissociation constant (K_D) that is at least 0.1 (e.g., at least 0.15, 0.175, 0.2, 0.25, 0.275, 0.3, 0.325, 0.35, 0.375, 0.4, 0.425, 0.45, 0.475, 0.5, 0.525, 0.55, 0.575, 0.6, 0.625, 0.65, 0.675, 0.7, 0.725, 0.75, 0.775, 0.8, 0.825, 0.85, 0.875, 0.9, 0.925, 0.95, or 0.975) nM. In some embodiments, the K_D of the anti-C5 antibody, or antigen binding fragment thereof, is no greater than 1 (e.g., no greater than 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, or 0.2) nM.

In other embodiments, the [$(K_D$ of the antibody for C5 at pH 6.0 at 25 C)/(K_D of the antibody for C5 at pH 7.4 at 25

C)] is greater than 21 (e.g., greater than 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500 or 8000).

Methods for determining whether an antibody binds to a protein antigen and/or the affinity for an antibody to a protein antigen are known in the art. The binding of an antibody to a protein antigen, for example, can be detected and/or quantified using a variety of techniques such as, but not limited to, Western blot, dot blot, surface plasmon resonance (SPR) detection (e.g., BIAcore system; Pharmacia Biosensor AB, Uppsala, Sweden and Piscataway, N.J.), or enzyme-linked immunosorbent assay (ELISA; Benny K. C. Lo (2004) "Antibody Engineering: Methods and Protocols," Humana Press (ISBN: 1588290921); John E. B. et al., *J. Immunol. Meth.*, 160:191-8, 1993; Jönsson, U. et al., *Ann. Biol. Clin.*, 51:19-26, 1993; Jönsson, U. et al., *Biotechniques*, 11:620-7, 1991). In addition, methods for measuring the affinity (e.g., dissociation and association constants) are set forth in the working examples.

As used herein, the term " k_a " refers to the rate constant for association of an antibody to an antigen. The term " k_d " refers to the rate constant for dissociation of an antibody from the antibody/antigen complex. And the term " K_D " refers to the equilibrium dissociation constant of an antibody-antigen interaction. The equilibrium dissociation constant is deduced from the ratio of the kinetic rate constants, $K_D=k_d/k_a$. Such determinations can be measured, for example, at 25 C or 37 C (see the working examples). The kinetics of antibody binding to human C5 can be determined, for example, at pH 8.0, 7.4, 7.0, 6.5 and 6.0 via SPR on a BIAcore 3000 instrument using an anti-Fc capture method to immobilize the antibody.

In one embodiment, the anti-C5 antibody, or antigen binding fragment thereof, blocks the cleavage of C5 into C5a and C5b. Through this blocking effect, for example, the pro-inflammatory effects of C5a and the generation of the C5b-9 membrane attack complex (MAC) at the surface of a cell are inhibited.

Methods for determining whether a particular antibody described herein inhibits C5 cleavage are known in the art. Inhibition of human complement component C5 can reduce the cell-lysing ability of complement in a subject's body fluids. Such reductions of the cell-lysing ability of complement present in the body fluid(s) can be measured by methods known in the art such as, for example, by a conventional hemolytic assay such as the hemolysis assay (Kabat and Mayer (eds.), "Experimental Immunology, 2nd Edition," 135-240, Springfield, IL, CC Thomas (1961), pages 135-139), or a conventional variation of that assay such as the chicken erythrocyte hemolysis method (Hillmen, P. et al., *N. Engl. J. Med.*, 350:552-9, 2004). Methods for determining whether a candidate compound inhibits the cleavage of human C5 into forms C5a and C5b are known in the art (Evans, M. et al., *Mol. Immunol.*, 32:1183-95, 1995). The concentration and/or physiologic activity of C5a and C5b in a body fluid can be measured, for example, by methods known in the art. For C5b, hemolytic assays or assays for soluble C5b-9 as discussed herein can be used. Other assays known in the art can also be used. Using assays of these or other suitable types, candidate agents capable of inhibiting human complement component C5 can be screened.

Immunological techniques such as, but not limited to, ELISA can be used to measure the protein concentration of C5 and/or its split products to determine the ability of an anti-C5 antibody, or antigen binding fragment thereof, to inhibit conversion of C5 into biologically active products. In some embodiments, C5a generation is measured. In some embodiments, C5b-9 neopeptide-specific antibodies are used to detect MAC formation.

Hemolytic assays can be used to determine the inhibitory activity of an anti-C5 antibody, or antigen binding fragment thereof, on complement activation. To determine the effect of an anti-C5 antibody, or antigen binding fragment thereof, on classical complement pathway-mediated hemolysis in a serum test solution in vitro, for example, sheep erythrocytes coated with hemolysin or chicken erythrocytes sensitized with anti-chicken erythrocyte antibody are used as target cells. The percentage of lysis is normalized by considering 100% lysis equal to the lysis occurring in the absence of the inhibitor. In some embodiments, the classical complement pathway is activated by a human IgM antibody, for example, as utilized in the Wieslab® Classical Pathway Complement Kit (Wieslab® COMPL CP310, Euro-Diagnostica, Sweden). Briefly, the test serum is incubated with an anti-C5 antibody, or antigen binding fragment thereof, in the presence of a human IgM antibody. The amount of C5b-9 that is generated is measured by contacting the mixture with an enzyme conjugated anti-C5b-9 antibody and a fluorogenic substrate and measuring the absorbance at the appropriate wavelength. As a control, the test serum is incubated in the absence of the anti-C5 antibody, or antigen binding fragment thereof. In some embodiments, the test serum is a C5-deficient serum reconstituted with a C5 polypeptide.

To determine the effect of an anti-C5 antibody, or antigen binding fragment thereof, on alternative pathway-mediated hemolysis, unsensitized rabbit or guinea pig erythrocytes can be used as the target cells. In some embodiments, the serum test solution is a C5-deficient serum reconstituted with a C5 polypeptide. The percentage of lysis is normalized by considering 100% lysis equal to the lysis occurring in the absence of the inhibitor. In some embodiments, the alternative complement pathway is activated by lipopolysaccharide molecules, for example, as utilized in the Wieslab® Alternative Pathway Complement Kit (Wieslab® COMPL AP330, Euro-Diagnostica, Sweden). Briefly, the test serum is incubated with an anti-C5 antibody, or antigen binding fragment thereof, in the presence of lipopolysaccharide. The amount of C5b-9 that is generated is measured by contacting the mixture with an enzyme conjugated anti-C5b-9 antibody and a fluorogenic substrate and measuring the fluorescence at the appropriate wavelength. As a control, the test serum is incubated in the absence of the anti-C5 antibody, or antigen binding fragment thereof.

In some embodiments, C5 activity, or inhibition thereof, is quantified using a CH50eq assay. The CH50eq assay is a method for measuring the total classical complement activity in serum. This test is a lytic assay, which uses antibody-sensitized erythrocytes as the activator of the classical complement pathway and various dilutions of the test serum to determine the amount required to give 50% lysis (CH50). The percent hemolysis can be determined, for example, using a spectrophotometer. The CH50eq assay provides an indirect measure of terminal complement complex (TCC) formation, since the TCC themselves are directly responsible for the hemolysis that is measured. The assay is known and commonly practiced by those of skill in the art. Briefly, to activate the classical complement pathway, undiluted serum samples (e.g., reconstituted human serum samples)

are added to microassay wells containing the antibody-sensitized erythrocytes to thereby generate TCC. Next, the activated sera are diluted in microassay wells, which are coated with a capture reagent (e.g., an antibody that binds to one or more components of the TCC). The TCC present in the activated samples bind to the monoclonal antibodies coating the surface of the microassay wells. The wells are washed and to each well is added a detection reagent that is detectably labeled and recognizes the bound TCC. The detectable label can be, e.g., a fluorescent label or an enzymatic label. The assay results are expressed in CH50 unit equivalents per milliliter (CH50 U Eq/mL).

Inhibition, e.g., as it pertains to terminal complement activity, includes at least a 5 (e.g., at least a 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55 or 60) % decrease in the activity of terminal complement in, e.g., a hemolytic assay or CH50eq assay as compared to the effect of a control antibody (or antigen-binding fragment thereof) under similar conditions and at an equimolar concentration. Substantial inhibition, as used herein, refers to inhibition of a given activity (e.g., terminal complement activity) of at least 40 (e.g., at least 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, or 95 or greater) %. In some embodiments, an anti-C5 antibody described herein contains one or more amino acid substitutions relative to the CDRs of eculizumab (i.e., SEQ ID NOs:1-6), yet retains at least 30 (e.g., at least 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 55, 60, 65, 70, 75, 80, 85, 90 or 95) % of the complement inhibitory activity of eculizumab in a hemolytic assay or CH50eq assay.

An anti-C5 antibody described herein has a serum half-life in humans that is at least 20 (e.g., at least 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54 or 55) days. In another embodiment, the anti-C5 antibody described herein has a serum half-life in humans that is at least 40 days. In another embodiment, the anti-C5 antibody described herein has a serum half-life in humans that is approximately 43 days. In another embodiment, the anti-C5 antibody described herein has a serum half-life in humans that is between 39-48 days. Methods for measuring the serum half-life of an antibody are known in the art. In some embodiments, an anti-C5 antibody, or antigen binding fragment thereof, described herein has a serum half-life that is at least 20 (e.g., at least 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 125, 150, 175, 200, 250, 300, 400 or 500) % greater than the serum half-life of eculizumab, e.g., as measured in one of the mouse model systems described in the working examples (e.g., the C5-deficient/NOD/scid mouse or hFcRn transgenic mouse model system).

In one embodiment, the antibody competes for binding with, and/or binds to the same epitope on C5 as an antibody described herein. The term "binds to the same epitope" with reference to two or more antibodies means that the antibodies bind to the same segment of amino acid residues, as determined by a given method. Techniques for determining whether antibodies bind to the same epitope on C5 with an antibody described herein include, for example, epitope mapping methods, such as, x-ray analyses of crystals of antigen:antibody complexes, and hydrogen/deuterium exchange mass spectrometry (HDX-MS). Other methods monitor the binding of the antibody to peptide antigen fragments or mutated variations of the antigen where loss of binding due to a modification of an amino acid residue within the antigen sequence is often considered an indication of an epitope component. In addition, computational combinatorial methods for epitope mapping can also be used.

These methods rely on the ability of the antibody of interest to affinity isolate specific short peptides from combinatorial phage display peptide libraries. Antibodies having the same VH and VL or the same CDR1, CDR2 and CDR3 sequences are expected to bind to the same epitope.

Antibodies that "compete with another antibody for binding to a target" refer to antibodies that inhibit (partially or completely) the binding of the other antibody to the target. Whether two antibodies compete with each other for binding to a target, i.e., whether and to what extent one antibody inhibits the binding of the other antibody to a target, may be determined using known competition experiments. In certain embodiments, an antibody competes with, and inhibits binding of another antibody to a target by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or 100%. The level of inhibition or competition may be different depending on which antibody is the "blocking antibody" (i.e., the antibody that is incubated first with the target). Competing antibodies can bind to, for example, the same epitope, an overlapping epitope or to adjacent epitopes (e.g., as evidenced by steric hindrance).

Anti-C5 antibodies, or antigen-binding fragments thereof described herein, used in the methods described herein can be generated using a variety of art-recognized techniques. Monoclonal antibodies can be obtained by various techniques familiar to those skilled in the art. Briefly, spleen cells from an animal immunized with a desired antigen are immortalized, commonly by fusion with a myeloma cell (Köhler, G. & Milstein, C., *Eur. J. Immunol.*, 6:511-9, 1976). Methods of immortalization include transformation with Epstein Barr Virus, oncogenes, or retroviruses or other methods known in the art. Colonies arising from single immortalized cells are screened for production of antibodies of the desired specificity and affinity for the antigen, and yield of the monoclonal antibodies produced by such cells may be enhanced by various techniques, including injection into the peritoneal cavity of a vertebrate host. Alternatively, one may isolate DNA sequences that encode a monoclonal antibody or a binding fragment thereof by screening a DNA library from human B cells (Huse, W. et al., *Science*, 246:1275-81, 1989).

III. Compositions

Also provided herein are compositions comprising an anti-C5 antibody or antigen binding fragment thereof. In one embodiment, the composition comprises an anti-C5 antibody comprising the CDR1, CDR2 and CDR3 domains in a heavy chain variable region having the sequence set forth in SEQ ID NO:12, and the CDR1, CDR2 and CDR3 domains in a light chain variable region having the sequence set forth in SEQ ID NO:8. In another embodiment, the anti-C5 antibody comprises heavy and light chains having the sequences shown in SEQ ID NOs:14 and 11, respectively. In another embodiment, the anti-C5 antibody comprises heavy and light chains having the sequences shown in SEQ ID NOs:20 and 11, respectively.

The compositions can be formulated as a pharmaceutical solution, e.g., for administration to a subject for the treatment or prevention of a complement-associated disorder. The pharmaceutical compositions generally include a pharmaceutically acceptable carrier. As used herein, a "pharmaceutically acceptable carrier" refers to, and includes, any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like that are physiologically compatible. The compositions can include a pharmaceutically acceptable salt,

e.g., an acid addition salt or a base addition salt, sugars, carbohydrates, polyols and/or tonicity modifiers.

The compositions can be formulated according to standard methods. Pharmaceutical formulation is an established art (see, for example, Gennaro (2000) "Remington: The Science and Practice of Pharmacy," 20th Edition, Lippincott, Williams & Wilkins (ISBN: 0683306472); Ansel et al. (1999) "Pharmaceutical Dosage Forms and Drug Delivery Systems," 7th Edition, Lippincott Williams & Wilkins Publishers (ISBN: 0683305727); and Kibbe (2000) "Handbook of Pharmaceutical Excipients American Pharmaceutical Association," 3rd Edition (ISBN: 091733096X)). In some embodiments, a composition can be formulated, for example, as a buffered solution at a suitable concentration and suitable for storage at 2-8 C (e.g., 4 C). In some embodiments, a composition can be formulated for storage at a temperature below 0C (e.g., -20 C or -80 C). In some embodiments, the composition can be formulated for storage for up to 2 years (e.g., 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 1 year, 1½ years or 2 years) at 2-8 C (e.g., 4 C). Thus, in some embodiments, the compositions described herein are stable in storage for at least 1 year at 2-8 C (e.g., 4 C).

The pharmaceutical compositions can be in a variety of forms. These forms include, e.g., liquid, semi-solid and solid dosage forms, such as liquid solutions (e.g., injectable and infusible solutions), dispersions or suspensions, tablets, pills, powders, liposomes and suppositories. The preferred form depends, in part, on the intended mode of administration and therapeutic application. Compositions containing a composition intended for systemic or local delivery, for example, can be in the form of injectable or infusible solutions. Accordingly, the compositions can be formulated for administration by a parenteral mode (e.g., intravenous, subcutaneous, intraperitoneal, or intramuscular injection). "Parenteral administration," "administered parenterally" and other grammatically equivalent phrases, as used herein, refer to modes of administration other than enteral and topical administration, usually by injection, and include, without limitation, intravenous, intranasal, intraocular, pulmonary, intramuscular, intraarterial, intrathecal, intracapsular, intraorbital, intracardiac, intradermal, intrapulmonary, intraperitoneal, transtracheal, subcutaneous, subcuticular, intraarticular, subcapsular, subarachnoid, intraspinal, epidural, intracerebral, intracranial, intracarotid and intrasternal injection and infusion.

IV. Outcomes

Provided herein are methods for treating PNH in a patient comprising administering to the patient an anti-C5 antibody. Symptoms of PNH include, but are not limited to, pallor, fatigue (e.g., tiredness, difficulty performing daily activities, trouble concentrating, dizziness, weakness), pain (e.g., stomach pain, leg pain or swelling, chest pain, back pain), dark-colored urine, shortness of breath, difficulty swallowing, yellowing of the skin and/or eyes, anemia, cytopenia, erectile dysfunction, blood clots, kidney disease, damage to organs, stroke or heart attack.

Patients treated according to the methods disclosed herein experience improvement in at least one sign of PNH. The treatment may produce at least one therapeutic effect selected from the group consisting of, for example, a reduction or cessation in pallor, fatigue, jaundice, anemia, cytopenia, abdominal pain, dyspnea, dysphagia, chest pain or erectile dysfunction.

In one embodiment, improvement is measured by terminal complement inhibition.

In another embodiment, lactate dehydrogenase (LDH) levels can be used to evaluate responsiveness to a therapy (e.g., a reduction of hemolysis as assessed by lactate dehydrogenase (LDH) levels is indicative of an improvement in at least one sign of PNH). LDH is a marker of intravascular hemolysis (Hill, A. et al., *Br. J. Haematol.*, 149:414-25, 2010; Hillmen, P. et al., *N. Engl. J. Med.*, 350:552-9, 2004; Parker, C. et al., *Blood*, 106:3699-709, 2005). Red blood cells contain large amounts of LDH, and a correlation between cell-free hemoglobin and LDH concentration has been reported in vitro (Van Lente, F. et al., *Clin. Chem.*, 27:1453-5, 1981) and in vivo (Kato, G. et al., *Blood*, 107:2279-85, 2006). The consequences of hemolysis are independent of anemia (Hill, A. et al., *Haematologica*, 93(s1):359Abs.0903, 2008; Kanakura, Y et al., *Int. J. Hematol.*, 93:36-46, 2011). LDH concentration obtained at baseline and then serially throughout a treatment period, is an important measure of hemolysis. Baseline levels of cell-free plasma hemoglobin are highly elevated in patients with PNH with LDH $\geq$ 1.5-fold above the upper limit of normal (LDH $\geq$ 1.5 $\times$ ULN), with a significant correlation between LDH and cell-free plasma hemoglobin (Hillmen, P. et al., *N. Engl. J. Med.*, 355:1233-43, 2006). The normal LDH value range is 105-333 IU/L (international units per liter).

LDH levels can be measured using any suitable test or assay, such as those described by Ferri F F, ed. *Ferri's Clinical Advisor* 2014. Philadelphia: Pa: Elsevier Mosby; 2014: Section IV—Laboratory tests and interpretation of results. LDH concentration can be measured in various samples obtained from a patient, in particular, serum samples. As used herein, the term "sample" refers to biological material from a subject. Although serum LDH concentration is of interest, samples can be derived from other sources, including, for example, single cells, multiple cells, tissues, tumors, biological fluids, biological molecules or supernatants or extracts of any of the foregoing. Examples include tissue removed for biopsy, tissue removed during resection, blood, urine, lymph tissue, lymph fluid, cerebrospinal fluid, mucous and stool samples. The sample used can vary based on the assay format, the detection method and the nature of the tumors, tissues, cells or extracts to be assayed. Methods for preparing samples are known in the art and can be readily adapted to obtain a sample that is compatible with the method utilized.

In one embodiment, patients treated according to the disclosed methods experience reductions in LDH levels to normal levels or to within 10%, or within 20% above what is considered the normal level (e.g., within 105-333 IU/L). In one embodiment, the patient's LDH levels are $\geq$ 1.5 fold above the upper limit of normal (LDH $\geq$ 1.5 $\times$ ULN) prior to initiating treatment.

In another embodiment, the treatment produces a shift toward normal levels of a hemolysis-related hematologic biomarker selected from the group consisting of free hemoglobin, haptoglobin, reticulocyte count, PNH red blood cell (RBC) clone and D-dimer.

In another embodiment, the treatment produces a reduction in the need for blood transfusions compared to baseline.

In another embodiment, the treatment produces a reduction in major adverse vascular events (MAVEs; e.g., thrombophlebitis/deep vein thrombosis, pulmonary embolus, myocardial infarction, transient ischemic attack, unstable angina, renal vein thrombosis/renal artery thrombosis/glomerular thrombosis, renal infarction, acute peripheral vascular occlusion, mesenteric/visceral vein/arterial thrombosis or infarction, hepatic/portal vein thrombosis, cerebral arterial occlusion/cerebrovascular accident, cerebral venous occlusion, renal arterial thrombosis, or multi-infarct dementia).

In another embodiment, the treatment produces a shift toward normal levels of a chronic disease associated biomarker selected from the group consisting of estimated glomerular filtration rate (eGFR) and spot urine:albumin:creatinine and plasma brain natriuretic peptide (BNP).

In another embodiment, the treatment produces a change from baseline in quality of life as assessed via the Functional Assessment of Chronic Illness Therapy (FACIT)-Fatigue Scale, version 4 and the European Organisation for Research and Treatment of Cancer, Quality of Life Questionnaire-Core 30 Scale compared to baseline.

V. Kits and Unit Dosage Forms

Also provided herein are kits that include a pharmaceutical composition containing an anti-C5 antibody or antigen binding fragment thereof, such as ravulizumab or BNJ421, and a pharmaceutically acceptable carrier, in a therapeutically effective amount adapted for use in the preceding methods. The kits optionally also can include instructions, e.g., comprising administration schedules, to allow a practitioner (e.g., a physician, nurse, or patient) to administer the composition contained therein to administer the composition to a patient having PNH. The kit also can include a syringe.

Optionally, the kits include multiple packages of the single-dose pharmaceutical compositions each containing an effective amount of the anti-C5 antibody, or antigen binding fragment thereof, for a single administration in accordance with the methods provided above. Instruments or devices necessary for administering the pharmaceutical composition(s) also may be included in the kits. For instance, a kit may provide one or more pre-filled syringes containing an amount of the anti-C5 antibody or antigen binding fragment thereof.

In one embodiment, a kit for treating PNH in a human pediatric patient comprises: (a) a dose of an anti-C5 antibody or antigen binding fragment thereof, comprising CDR1, CDR2 and CDR3 domains of the heavy chain variable region having the sequence set forth in SEQ ID NO:12, and CDR1, CDR2 and CDR3 domains of the light chain variable region having the sequence set forth in SEQ ID NO:8; and (b) instructions for using the anti-C5 antibody or antigen binding fragment thereof, according to any of the methods described herein.

In one embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody or antigen binding fragment thereof, is administered to a patient weighing $\geq$ 5 to $<$ 10 kg: (a) once on Day 1 at a dose of 600 mg; and (b) on Day 15 and every four weeks thereafter at a dose of 300 mg.

In another embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody or antigen binding fragment thereof, is administered to a patient weighing ≥ 10 to < 20 kg: (a) once on Day 1 at a dose of 600 mg; and (b) on Day 15 and every four weeks thereafter at a dose of 600 mg.

In another embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody, or antigen binding fragment thereof, is administered to a patient weighing 20 to < 30 kg: (a) once on Day 1 at a dose of 900 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 2100 mg.

In another embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody, or antigen binding fragment thereof, is administered to a patient weighing 30 to < 40 kg: (a) once on Day 1 at a dose of 1200 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 2700 mg.

In another embodiment, the kit comprises a dose of an anti-C5 antibody, or antigen binding fragment thereof, wherein the anti-C5 antibody, or antigen binding fragment thereof, is administered to a patient weighing 40 to < 60 kg: (a) once on Day 1 at a dose of 2400 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3000 mg.

In another embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody, or antigen binding fragment thereof, is administered to a patient weighing 60 to < 100 kg: (a) once on Day 1 at a dose of 2700 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3300 mg. In another embodiment, the kit comprises a dose of an anti-C5 antibody or antigen binding fragment thereof, wherein the anti-C5 antibody, or antigen binding fragment thereof, is administered to a patient weighing 100 kg: (a) once on Day 1 at a dose of 3000 mg; and (b) on Day 15 and every eight weeks thereafter at a dose of 3600 mg.

The following examples are merely illustrative and should not be construed as limiting the scope of this disclosure in any way as many variations and equivalents will become apparent to those skilled in the art upon reading the present disclosure. The contents of all references, Genbank entries, patents and published patent applications cited throughout this application are expressly incorporated herein by reference.

EXAMPLES

Example 1: A Phase 3, Open-Label Study of Ravulizumab in Children and Adolescents with Paroxysmal Nocturnal Hemoglobinuria (PNH)

In this Phase 3, open-label study, the pharmacokinetic(s) (PK), pharmacodynamics (PD), efficacy, and safety of ravulizumab are assessed in pediatric patients with PNH.

1. Objectives

The primary objective of the study is to assess the PK, PD, safety and efficacy of ravulizumab in pediatric patients with PNH.

2. Endpoints

The primary endpoints are PK/PD parameters (trough and peak) at Baseline and Weeks 2, 10, 18, and 26. PK param-

eters include maximum serum concentration (C_{max}), trough serum concentration (measured at end of dosing interval at steady state; C_{trough}), and accumulation ratio. PD parameters include measuring the change in free C5 concentrations and in chicken red blood cell (cRBC) hemolytic activity over time.

Secondary endpoints include assessment of the following parameters:

Percentage change in LDH from baseline to Day 183 (Week 26);

Transfusion avoidance (TA), defined as the proportion of patients who remain transfusion-free and do not require a transfusion through Day 183 (Week 26);

Change in quality of life (QoL), as measured by Pediatric Functional Assessment of Chronic Therapy (FACIT) Fatigue questionnaire (patients ≥ 5 years of age), from baseline to Day 183 (Week 26);

Proportion of patients with stabilized hemoglobin, defined as avoidance of a ≥ 2 g/dL decrease in hemoglobin level from baseline in the absence of transfusion through Day 183 (Week 26);

Percentage change in free hemoglobin from baseline to Day 183 (Week 26);

Proportion of patients with breakthrough hemolysis, defined as at least one new or worsening symptom or sign of intravascular hemolysis (fatigue, hemoglobinuria, abdominal pain, shortness of breath [dyspnea], anemia, major adverse vascular event [MAVE, including thrombosis], dysphagia, or erectile dysfunction) in the presence of elevated LDH as follows: for patients who enter the study naïve to complement inhibitor treatment, elevated $LDH \geq 2 \times$ the upper limit of normal (ULN) after prior LDH reduction to $< 1.5 \times$ ULN on therapy; for patients who enter the study stabilized on eculizumab treatment, elevated $LDH \geq 2 \times$ ULN.

With respect to safety endpoints, the safety and tolerability of ravulizumab is evaluated from baseline to Week 26 and throughout the extension period by physical examinations, vital signs, physical growth (height, weight, and head circumference [the latter only in patients who are ≤ 3 years of age]), electrocardiograms (ECGs), laboratory assessments, and incidence of adverse events (AEs) and serious adverse events (SAEs). The proportion of patients who develop antidrug antibodies (ADAs) is also assessed.

3. Study Design

This is a Phase 3, open-label, single-arm multicenter study to evaluate the PK/PD, safety, and efficacy of ravulizumab administered by intravenous (IV) infusion to pediatric patients (< 18 years of age) with PNH. The study consists of a 4-week Screening Period, a 26-week Primary Evaluation Period and an Extension Period.

Consenting patients are screened for study eligibility up to 4 weeks prior to Day 1. Patients who satisfy all of the inclusion criteria and all of the exclusion criteria are enrolled into the Primary Evaluation Period and receive a weight-based loading dose of ravulizumab on Day 1, followed by weight-based maintenance treatment with ravulizumab on Day 15 and once every 8 weeks (q8w) thereafter for patients weighing ≥ 20 kg, or once every 4 weeks (q4w) for patients weighing < 20 kg, for a total of 26 weeks of treatment. For

patients entering the study on eculizumab therapy, Day 1 of study treatment occurs 2 weeks from the patient's last dose of eculizumab.

An interim analysis of data, including ravulizumab PK and free C5 levels, is conducted after 4 patients weighing ≥ 5 kg to <40 kg have completed dosing through Day 71. Enrollment of patients proceeds without interruption while the analysis is ongoing. The accrued safety and PK/PD data is assessed to ensure that ravulizumab treatment is well tolerated and is providing adequate complement inhibition. In addition, an independent Data Monitoring Committee (DMC) reviews safety data from the study on a regular basis.

After completion of all assessments on Day 183, all patients enter an Extension Period and continue to receive ravulizumab according to the appropriate weight-based regimen. The Extension Period continues until the product is registered or approved (in accordance with country-specific regulations) or for up to 2 years, whichever occurs first. The end of trial is defined as the last patient's last visit in the Extension Period.

4. Schedule of Assessments

The Schedule of Assessments is set forth in Table 1 for the Screening and Primary Evaluation Period and in Table 2 for the Extension Period.

Additional (unscheduled) visits outside the specified visits are permitted at the discretion of the Investigator. Procedures, tests, and assessments are performed at the discretion of the Investigator. All tests, procedures, or assessments performed at the Unscheduled Visits are recorded on the electronic case report forms (eCRFs).

Additionally, if a suspected event of breakthrough hemolysis occurs, LDH, PK and PD parameters are analyzed at the central laboratory. If the suspected event of breakthrough does not occur at a scheduled visit, an unscheduled visit occurs for evaluation of the patient and collection of the required LDH, PK and PD parameters. For purposes of defining breakthrough hemolysis, assessment of LDH is based on a central laboratory value.

TABLE 1

Schedule of Study Visits and Assessments: Screening Through End of Primary Evaluation Period											
	Period										
	Screening		Initial Evaluation Period								
			Study Day								
	-28 to -1	1	15	43	71	99	127	155	183 ^a /ET	Window (day)	
	N/A		± 3	± 3	± 3	± 5	± 5	± 5	± 2		
Informed consent	X										
Confirmation or administration of meningococcal vaccination ^b	X	X									
Confirmation of <i>H influenza</i> type B and <i>S pneumoniae</i> vaccination (per local/national guidelines)	X										
Medical history and demographics	X										
PNH clone size ^c	X	X			X						
Head circumference (patients $\leq$ 3 years of age only)	X	X	X	X	X	X	X	X	X		
Height and weight ^d	X	X	X	X	X	X	X	X	X		
Pregnancy test ^e	X	X	X		X		X		X		
Record transfusions (during and between visits)	X	X	X	X	X	X	X	X	X		
PNH symptomatology ^f	X	X	X	X	X	X	X	X	X		
Pediatric FACIT-Fatigue questionnaire ^g		X	X		X		X		X		
Physical examination	X										
Abbreviated physical examination ^h		X	X	X	X	X	X	X			
Vital signs ⁱ	X	X	X	X	X	X	X	X	X		
Safety 12-Lead ECG ^j	X				X						
Chemistry including LDH ^k	X	X	X	X	X	X	X	X	X		
Hematology including free hemoglobin and coagulation ^k	X	X	X	X	X	X	X	X	X		
Urinalysis and urine chemistry ^k	X	X	X		X		X		X		
PK/PD sampling ^m		X	X		X		X		X		
Immunogenicity (ADA) ⁿ		X			X		X		X		
Review safety card ^o		X	X	X	X	X	X	X	X		
Breakthrough hemolysis ^p										←Monitor continuously→	
Concomitant medications	X									←Monitor continuously→	
Adverse events	X									←Monitor continuously→	
ravulizumab administration (patients weighing < 20 kg) ^{q, r}		X ^s	X	X	X	X	X	X			

TABLE 1-continued

Schedule of Study Visits and Assessments: Screening Through End of Primary Evaluation Period									
Period									
Screening		Initial Evaluation Period							
		Study Day							
-28 to -1		1	15	43	71	99	127	155	183 ^a /ET
		Window (day)							
N/A		±3	±3	±3	±5	±5	±5	±5	±2
ravulizumab administration (patients weighing ≥ 20 kg) ^g		X ^s	X		X		X		

Abbreviations:

ADA = antidrug antibody;

ECG = electrocardiogram;

eCRF = electronic case report form;

EOI = end of infusion;

ET = early termination;

FACIT-Fatigue = Functional Assessment of Chronic Illness Therapy-Fatigue Scale;

LDH = lactate dehydrogenase;

N/A = not applicable;

PD = pharmacodynamics;

PK = pharmacokinetics;

PNH = paroxysmal nocturnal hemoglobinuria;

RBC = red blood cell;

WBC = white blood cell

^aThe primary endpoint assessment is before dosing on Day 183. Dosing on Day 183 is the start of the Extension Period. Please refer to additional Day 183 post-dose assessments in Table 2.^bAll patients are vaccinated against meningococcal infections within 3 years prior to, or at the time of, initiating study drug. Patients who initiate study drug treatment less than 2 weeks after receiving a meningococcal vaccine receive treatment with appropriate prophylactic antibiotics until 2 weeks after vaccination. Patients who have not been vaccinated prior to initiating ravulizumab treatment receive prophylactic antibiotics prior to and for at least 2 weeks after meningococcal vaccination. Patients who cannot be vaccinated receive antibiotic prophylaxis for the entire treatment period and for 8 months following last dose.^cWBC (granulocyte and monocyte) and RBC clone size measured by high-sensitivity flow cytometry at Screening and Day 1; RBC clone size only on Day 71 and Day 183.^dHeight at baseline and Day 183 only. Weight is collected pre-dose on dosing days.^ePregnancy testing is required only for female patients of childbearing potential (i.e., have achieved menarche). Serum pregnancy test is performed at Screening and Day 183, urine pregnancy test at all other required time points. A negative pregnancy test result is required prior to administering study drug to female patients of childbearing potential at the indicated study visits.^fInvestigator or designee assessment of the following events: fatigue, hemoglobinuria abdominal pain, dyspnea, dysphagia, chest pain, and erectile dysfunction. On dosing days, assessments are performed prior to dosing.^gOn dosing days, assessments are performed prior to dosing. Pediatric FACIT-Fatigue only in patients ≥5 years of age (self-reported by patients who were ≥8 years of age at the time of enrollment and reported by caregivers for patients who were ≥5 to <8 years of age at the time of enrollment).^hAbbreviated physical examination consists of a body system relevant examination based upon Investigator (or qualified designee) judgment and patient symptoms. At least one body system is checked for an abbreviated exam.ⁱVital sign measurements are taken after the patient has been resting for at least 5 minutes and include systolic and diastolic blood pressure (BP) (millimeters of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (degrees Celsius [°C.] or degrees Fahrenheit [°F.]). On dosing days, vital signs are taken pre-dose.^jSingle 12-lead ECG is collected at Screening and pre-dose on Day 71 and Day 183. Patients are supine for approximately 5 to 10 minutes before ECG collection and remain supine (but awake) during ECG collection.^kClinical laboratory measurements are collected pre-dose on dosing days and not from a heparinized line.^lFor patients entering the study on ecilizumab therapy, screening LDH is obtained within 24 hours prior to a scheduled ecilizumab dose.^mSerum samples for PK/PD analyses are collected at the indicated visits. For indicated visits falling on dosing days, samples are collected pre-dose (within 0.5 hours prior to the start of infusion) and at end of infusion (EOI) (within 0.5 hours after the EOI from the patient's opposite, noninfused arm). To minimize needle sticks to the patient, the pre-dose sample is drawn through the venous access created for the dose infusion, prior to administration of the dose. As noted, the post-dose sample is drawn from the opposite, non-infused arm. For indicated visits not falling on dosing days, samples are collected at any time that visit day. If a supplemental dose is administered PK/PD samples are collected at pre-dose and at end of infusion. If a loading dose is administered as two separate infusions <24 hours apart, PK/PD samples are collected before the first infusion (i.e., the pre-dose sample) and after the second infusion (i.e., the end of infusion sample). All collection times are recorded in the eCRF.ⁿADA serum samples are collected pre-dose on Days 1, 71 and 127 or at any time during an ET visit. Day 183 collection occurs prior to first dose in the Extension Period.^oReview the Clinical Trial Participant Safety Information Card with the patient/caregiver and

discuss the importance of carrying the safety card at all times, and the risks ravulizumab treatment, including the risk of meningococcal infection.

^pIf a suspected event of breakthrough hemolysis occurs, LDH, PK and PD parameters are to be analyzed at the central laboratory. If the suspected event of breakthrough does not occur at a scheduled visit, an unscheduled visit occurs for evaluation of the patient and collection of the required LDH, PK and PD parameters.^qDose regimen is based on body weight obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.^rShould a patient's weight change from <20 kg to ≥20 kg on a "q4w only" visit, the patient receives the q4w dose that day. At the patient's next q8w visit, the new q8w dose is given.^sFor patients entering the study on ecilizumab therapy, Day 1 occurs 2 weeks from the patient's last dose of ecilizumab

Abbreviations: ADA=antidrug antibody; ECG=electrocardiogram; eCRF=electronic case report form; EOI=end of infusion; ET=early termination; FACIT-Fatigue=Functional Assessment of Chronic Illness Therapy-Fatigue Scale; LDH=lactate dehydrogenase; N/A=not applicable; PD=pharmacodynamics; PK=pharmacokinetics; PNH=paroxysmal nocturnal hemoglobinuria; RBC=red blood cell; WBC=white blood cell

^a The primary endpoint assessment is before dosing on Day 183. Dosing on Day 183 is the start of the Extension Period. Please refer to additional Day 183 post-dose assessments in Table 2.

^b All patients are vaccinated against meningococcal infections within 3 years prior to, or at the time of, initiating study drug. Patients who initiate study drug treatment less than 2 weeks after receiving a meningococcal vaccine receive treatment with appropriate prophylactic antibiotics until 2 weeks after vaccination. Patients who have not been vaccinated prior to initiating ravulizumab treatment receive prophylactic antibiotics prior to and for at least 2 weeks after meningococcal vaccination. Patients who cannot be vaccinated receive antibiotic prophylaxis for the entire treatment period and for 8 months following last dose.

^c WBC (granulocyte and monocyte) and RBC clone size measured by high-sensitivity flow cytometry at Screening and Day 1; RBC clone size only on Day 71 and Day 183.

^d Height at baseline and Day 183 only. Weight is collected pre-dose on dosing days.

^e Pregnancy testing is required only for female patients of childbearing potential (i.e., have achieved menarche). Serum pregnancy test is performed at Screening and Day 183, urine pregnancy test at all other required time points. A negative pregnancy test result is required prior to administering study drug to female patients of childbearing potential at the indicated study visits.

^f Investigator or designee assessment of the following events: fatigue, hemoglobinuria abdominal pain, dyspnea, dysphagia, chest pain, and erectile dysfunction. On dosing days, assessments are performed prior to dosing.

^g On dosing days, assessments are performed prior to dosing. Pediatric FACIT-Fatigue only in patients ≥ 5 years of age (self-reported by patients who were ≥ 8 years of age at the time of enrollment and reported by caregivers for patients who were ≥ 5 to < 8 years of age at the time of enrollment).

^h Abbreviated physical examination consists of a body system relevant examination based upon Investigator (or qualified designee) judgment and patient symptoms. At least one body system is checked for an abbreviated exam.

ⁱ Vital sign measurements are taken after the patient has been resting for at least 5 minutes and include systolic and diastolic blood pressure (BP) (millimeters of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (degrees Celsius [C] or degrees Fahrenheit [° F.]). On dosing days, vital signs are taken pre-dose.

^j Single 12-lead ECG is collected at Screening and pre-dose on Day 71 and Day 183. Patients are supine for approximately 5 to 10 minutes before ECG collection and remain supine (but awake) during ECG collection.

^k Clinical laboratory measurements are collected pre-dose on dosing days and not from a heparinized line.

^l For patients entering the study on eculizumab therapy, screening LDH is obtained within 24 hours prior to a scheduled eculizumab dose.

^m Serum samples for PK/PD analyses are collected at the indicated visits. For indicated visits falling on dosing days, samples are collected pre-dose (within 0.5 hours prior to the start of infusion) and at end of infusion (EOI) (within 0.5 hours after the EOI from the patient's opposite, noninfused arm). To minimize needle sticks to the patient, the pre-dose sample is drawn through the venous access created for the dose infusion, prior to administration of the dose. As noted, the post-dose sample is drawn from the opposite, non-infused arm. For indicated visits not falling on dosing days, samples are collected at any time that visit day. If a supplemental dose is administered PK/PD samples are collected at pre-dose and at end of infusion. If a loading dose is administered as two separate infusions < 24 hours apart, PK/PD samples are collected before the first infusion (i.e., the pre-dose sample) and after the second infusion (i.e., the end of infusion sample). All collection times are recorded in the eCRF.

ⁿ ADA serum samples are collected pre-dose on Days 1, 71 and 127 or at any time during an ET visit. Day 183 collection occurs prior to first dose in the Extension Period.

^o Review the Clinical Trial Participant Safety Information Card with the patient/caregiver and discuss the importance of carrying the safety card at all times, and the risks ravulizumab treatment, including the risk of meningococcal infection.

^p If a suspected event of breakthrough hemolysis occurs, LDH, PK and PD parameters are to be analyzed at the central laboratory. If the suspected event of breakthrough does not occur at a scheduled visit, an unscheduled visit occurs for evaluation of the patient and collection of the required LDH, PK and PD parameters.

^q Dose regimen is based on body weight obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.

^r Should a patient's weight change from < 20 kg to ≥ 20 kg on a "q4w only" visit, the patient receives the q4w dose that day. At the patient's next q8w visit, the new q8w dose is given.

^s For patients entering the study on eculizumab therapy, Day 1 occurs 2 weeks from the patient's last dose of eculizumab

TABLE 2

Schedule of Study Visits and Assessments: Extension Period														
	Study Day													
	183 ^a	211 q4w	239	267 q4w	295	323 q4w	351	379 q4w	407	435 q4w	463	491 q4w	519	547 q4w
	±2	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5
PNH clone size ^b								X						
Head circumference (patients ≤ 3 years of age)			X		X		X		X		X		X	
Height and weight ^c		X	X	X	X	X	X	X	X	X	X	X	X	X
Pregnancy test ^d			X		X		X							
Record transfusions			X		X		X							
PNH symptomatology ^e			X		X		X							
Pediatric FACIT-Fatigue questionnaire ^f							X							
Physical examination														
Abbreviated physical examination ^g			X		X		X							

TABLE 2-continued

Schedule of Study Visits and Assessments: Extension Period														
Vital signs ^h		X			X		X							
Safety 12-Lead ECG ⁱ														
Chemistry including LDH ^j		X			X		X							
Hematology including free hemoglobin and coagulation ^j		X			X		X							
Urinalysis and urine chemistry ^j		X			X		X							
PK/PD sampling ^k	X												X	
Immunogenicity (ADA) ^l													X	
Review safety card ^m	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Breakthrough hemolysis ⁿ														
Concomitant medications														
Adverse events														
ravulizumab administration (patients weighing <20 kg) ^{o, p}	X ^a	X	X	X	X	X	X	X	X	X	X	X	X	X
ravulizumab administration (patients weighing ≥20 kg) ^o	X ^a		X		X		X		X		X		X	
Study Day														
		575	603 q4w	631	659 q4w	687	715 q4w	743 q4w	771 q4w	799	827 q4w	855	883 q4w	911/ ET/ EOS
		Window (day)												
		±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5	±5
PNH clone size ^b		X												X
Head circumference (patients ≤ 3 years of age)		X		X		X		X		X		X		X
Height and weight ^c		X	X	X	X	X	X	X	X	X	X	X	X	X
Pregnancy test ^d		X						X						X
Record transfusions		X						X						X
PNH symptomatology ^e		X						X						X
Pediatric FACIT-Fatigue questionnaire ^f		X						X						X
Physical examination														X
Abbreviated physical examination ^g		X						X						
Vital signs ^h		X						X						X
Safety 12-Lead ECG ⁱ		X						X						X
Chemistry including LDH ^j		X						X						X
Hematology including		X						X						X

TABLE 2-continued

Schedule of Study Visits and Assessments: Extension Period													
free hemoglobin and coagulation ^j	X						X						X
Urinalysis and urine chemistry ^j	X						X						X
PK/PD sampling ^k													
Immunogenicity (ADA) ^l							X						
Review safety card ^m	X	X	X	X	X	X	X	X	X	X	X	X	X
Breakthrough hemolysis ⁿ							←Monitor continuously→						
Concomitant medications							←Monitor continuously→						
Adverse events							←Monitor continuously→						
ravulizumab administration (patients weighing <20 kg) ^{o, p}	X	X	X	X	X	X	X	X	X	X	X	X	X
ravulizumab administration (patients weighing ≥20 kg) ^o	X		X		X		X		X		X		

Abbreviations:

ADA = antidrug antibody;

ECG = electrocardiogram;

eCRF = electronic case report form;

EOS = end of study;

ET = early termination;

FACIT-Fatigue = Functional Assessment of Chronic Illness Therapy-Fatigue Scale;

LDH = lactate dehydrogenase;

PD = pharmacodynamics;

PK = pharmacokinetics;

PNH = paroxysmal nocturnal hemoglobinuria;

q4w = once every 4 weeks;

RBC = red blood cell

^aExtension Period begins at the start of Day 183 dosing.^bGranulocyte and RBC clone size measured by high-sensitivity flow cytometry.^cHeight at Days 575 and 911 only. Weight is collected pre-dose.^dPregnancy testing is required only for female patients of childbearing potential (i.e., have achieved menarche). Serum pregnancy test is performed at early termination (ET) only, unless required by site policy. Urine pregnancy test at all other required time points. A negative pregnancy test result is required prior to administering study drug to female patients of childbearing potential at the indicated study visits.^eInvestigator or designee assessment of the following events: fatigue, hemoglobinuria, abdominal pain, dyspnea, dysphagia, chest pain, and erectile dysfunction. On dosing days, assessments are performed prior to dosing.^fOn dosing days, assessments are performed prior to dosing. Pediatric FACIT-Fatigue only in patients ≥5 years of age (self-reported by patients who were ≥8 years of age at the time of enrollment and reported by caregivers for patients who were ≥5 to <8 years of age at the time of enrollment).^gAbbreviated physical examination consists of a body system relevant examination based upon Investigator (or qualified designee) judgment and patient symptoms. At least one body system is checked for an abbreviated exam.^hVital sign measurements are taken after the patient has been resting for at least 5 minutes and include systolic and diastolic BP (millimeters of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (degrees Celsius [C.] or degrees Fahrenheit [° F.]). On dosing days, vital signs are taken pre-dose.ⁱSingle 12-lead ECG is collected at Day 911 or ET. Patients are supine for approximately 5 to 10 minutes before ECG collection and remain supine (but awake) during ECG collection.^jClinical laboratory measurements are collected pre-dose and not from a heparinized line.^kSerum samples for PK/PD analyses are collected at end-of-infusion on Day 183. Pre-dose (within 0.5 hours prior to the start of infusion) and at end of infusion (within 0.5 hours after the end of infusion) on Days 351, 575, and 743; and at any time on Day 911 or early termination. To minimize needle sticks to the patient, the pre-dose sample is drawn through the venous access created for the dose infusion, prior to administration of the dose. End-of-infusion samples are drawn from the patient's opposite, non-infused arm. All collection times are recorded. In the event that supplemental dose is administered, pre-dose and end of infusion blood samples will be collected for PK/PD analysis. In the event of breakthrough hemolysis, a serum sample for PK/PD analysis is collected.^lSamples for ADA are collected pre-dose or at any time during the visit when a dose of a study drug is not administered.^mReview the Clinical Trial Participant Safety Information Card with the patient and discuss the importance of carrying the safety card at all times, and the risks ravulizumab treatment, including the risk of meningococcal infection.ⁿIf a suspected event of breakthrough hemolysis occurs, LDH, PK and PD parameters are analyzed at the central laboratory. If the suspected event of breakthrough does not occur at a scheduled visit, an unscheduled visit occurs for evaluation of the patient and collection of the required LDH, PK and PD parameters.^oDose regimen is based on body weight obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.^pShould a patient's weight change from <20 kg to ≥20 kg on a "q4w only" visit, the patient receives the q4w dose that day. At the patient's next q8w visit, the new q8w dose is given.

Abbreviations: ADA=antidrug antibody;

ECG=electrocardiogram; eCRF=electronic case report form; EOS=end of study; ET=early termination; FACIT-Fatigue=Functional Assessment of Chronic Illness Therapy-Fatigue Scale;

LDH=lactate dehydrogenase;

PD=pharmacodynamics; PK=pharmacokinetics;

PNH=paroxysmal nocturnal hemoglobinuria; q4w=once

every 4 weeks; RBC=red blood cell

^a Extension Period begins at the start of Day 183 dosing.^b Granulocyte and RBC clone size measured by high-sensitivity flow cytometry.^c Height at Days 575 and 911 only. Weight is collected pre-dose.^d Pregnancy testing is required only for female patients of childbearing potential (i.e., have achieved menarche). Serum pregnancy test is performed at early termination (ET) only, unless required by site policy. Urine pregnancy test at all other required time points. A negative pregnancy test result is

required prior to administering study drug to female patients of childbearing potential at the indicated study visits.

^e Investigator or designee assessment of the following events: fatigue, hemoglobinuria, abdominal pain, dyspnea, dysphagia, chest pain, and erectile dysfunction. On dosing days, assessments are performed prior to dosing.

^f On dosing days, assessments are performed prior to dosing. Pediatric FACIT-Fatigue only in patients ≥ 5 years of age (self-reported by patients who were ≥ 8 years of age at the time of enrollment and reported by caregivers for patients who were ≥ 5 to < 8 years of age at the time of enrollment).

^g Abbreviated physical examination consists of a body system relevant examination based upon Investigator (or qualified designee) judgment and patient symptoms. At least one body system is checked for an abbreviated exam.

^h Vital sign measurements are taken after the patient has been resting for at least 5 minutes and include systolic and diastolic BP (millimeters of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (degrees Celsius [C] or degrees Fahrenheit [° F.]). On dosing days, vital signs are taken pre-dose.

ⁱ Single 12-lead ECG is collected at Day 911 or ET. Patients are supine for approximately 5 to 10 minutes before ECG collection and remain supine (but awake) during ECG collection.

^j Clinical laboratory measurements are collected pre-dose and not from a heparinized line.

^k Serum samples for PK/PD analyses are collected at end-of-infusion on Day 183. Pre-dose (within 0.5 hours prior to the start of infusion) and at end-of-infusion (within 0.5 hours after the end of infusion) on Days 351, 575, and 743; and at any time on Day 911 or early termination. To minimize needle sticks to the patient, the pre-dose sample is drawn through the venous access created for the dose infusion, prior to administration of the dose. End-of-infusion samples are drawn from the patient's opposite, non-infused arm. All collection times are recorded. In the event that supplemental dose is administered, pre-dose and end of infusion blood samples will be collected for PK/PD analysis. In the event of breakthrough hemolysis, a serum sample for PK/PD analysis is collected.

^l Samples for ADA are collected pre-dose or at any time during the visit when a dose of a study drug is not administered.

^m Review the Clinical Trial Participant Safety Information Card with the patient and discuss the importance of carrying the safety card at all times, and the risks ravulizumab treatment, including the risk of meningococcal infection.

ⁿ If a suspected event of breakthrough hemolysis occurs, LDH, PK and PD parameters are analyzed at the central laboratory. If the suspected event of breakthrough does not occur at a scheduled visit, an unscheduled visit occurs for evaluation of the patient and collection of the required LDH, PK and PD parameters.

^o Dose regimen is based on body weight obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.

^p Should a patient's weight change from < 20 kg to ≥ 20 kg on a "q4w only" visit, the patient receives the q4w dose that day. At the patient's next q8w visit, the new q8w dose is given.

5. Study Population

Pediatric patients (age < 18 years) with a documented diagnosis of PNH are enrolled and treated with ravulizumab at approximately 40 investigative sites globally.

Approximately 12 patients are enrolled to ensure at least 10 evaluable patients complete the 26-week period. Individuals who do not meet the criteria for participation in this study (screen failure) can be rescreened once.

Patients are eligible for enrollment in the study only if they meet all of the following criteria and none of the exclusion criteria:

1. Male and female patients < 18 years of age and weighing ≥ 5 kg at the time of consent.
2. Documented diagnosis of PNH, confirmed by high-sensitivity flow cytometry evaluation (Borowitz, M. et al., *Cytometry B Clin. Cytom.*, 78:211-30, 2010) of red blood cells (RBCs) and white blood cells (WBCs), with granulocyte or monocyte clone size of $\geq 5\%$.
3. For patients not currently treated with eculizumab, presence of one or more of the following PNH-related signs or symptoms within three months of Screening: fatigue, hemoglobinuria, abdominal pain, shortness of breath (dyspnea), anemia, history of a major adverse vascular event (including thrombosis), dysphagia, or erectile dysfunction; or history of packed red blood cell (pRBC) transfusion due to PNH.

4. LDH values at Screening as follows:

- a. For patients not currently treated with eculizumab, LDH level $\geq 1.5 \times \text{ULN}$.
- b. For patients who are currently taking eculizumab, LDH $\leq 1.5 \times \text{ULN}$ (sample must be obtained on a scheduled eculizumab-dosing day prior to dose administration [i.e., at trough eculizumab level] and analyzed by the central laboratory).

5. To reduce the risk of meningococcal infection (*N. meningitidis*), all patients must be vaccinated against meningococcal infections within three years prior to, or at the time of, initiating study drug. Patients who initiate study drug treatment less than two weeks after receiving a meningococcal vaccine receive treatment with appropriate prophylactic antibiotics until two weeks after vaccination. Patients who cannot be vaccinated receive antibiotic prophylaxis for the entire treatment period and for eight months following last dose.

6. Patients must have been vaccinated against *Haemophilus influenzae* type b (Hib) and *Streptococcus pneumoniae* according to national and local vaccination schedule guidelines, as appropriate.

7. Female patients of childbearing potential (i.e., have achieved menarche) and male patients with female partners of childbearing potential follow protocol-specified guidance for avoiding pregnancy while on treatment and for 8 months after last dose of study drug. Patient's legal guardian gives written informed consent and the patient gives written informed assent and comply with the study visit schedule.

Patients are excluded from study enrollment if they meet any of the following criteria:

1. Platelet count $< 30,000/\text{mm}^3$ ($30 \times 10^9/\text{L}$) at Screening.
2. Absolute neutrophil count $< 500/\mu\text{L}$ ($0.5 \times 10^9/\text{L}$) at Screening.
3. History of bone marrow transplantation.
4. History of *N. meningitidis* infection.
5. History of unexplained, recurrent infection.
6. Active systemic bacterial, viral, or fungal infection within 14 days prior to study drug administration on Day 1.
7. History of malignancy within 5 years of Screening with the exception of adequately treated nonmelanoma skin cancer or carcinoma in situ of the cervix.
8. History of or ongoing major cardiac, pulmonary, renal, endocrine, or hepatic disease (e.g., active hepatitis) that, in the opinion of the Investigator or Sponsor, precludes the patient's participation in an investigational clinical trial.
9. Unstable medical conditions (e.g., myocardial ischemia, active gastrointestinal bleed, severe congestive heart failure, anticipated need for major surgery within 6 months of Screening, coexisting chronic anemia unrelated to PNH) that makes them unlikely to tolerate the requirements of the protocol.
10. Concomitant use of anticoagulants is prohibited if not on a stable regimen for at least 2 weeks prior to Day 1.
11. History of hypersensitivity to any ingredient contained in the study drug, including hypersensitivity to murine proteins.
12. Females who plan to become pregnant or are currently pregnant or breastfeeding.
13. Females of childbearing potential who have a positive pregnancy test result at Screening or on Day 1.

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14. Participation in another interventional treatment study or use of any experimental therapy within 30 days before initiation of study drug on Day 1 in this study or within 5 half-lives of that investigational product, whichever is greater.
15. Known or suspected history of drug or alcohol abuse or dependence within 1 year prior to the start of Screening.
16. Known medical or psychological condition(s) or risk factor that, in the opinion of the Investigator or Sponsor, might interfere with the patient's full participation in the study, pose any additional risk for the patient, or confound the assessment of the patient or outcome of the study.
6. Study Treatment
- Ravulizumab is a humanized, anti-C5 mAb produced in Chinese hamster cells. The ravulizumab drug product is

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not frozen. Study drug is stored in a secure, limited-access storage area, and the temperature is monitored daily.

The admixed drug product is at room temperature prior to administration. The material is not heated (e.g., by using a microwave or other heat source) other than by ambient air temperature.

Ravulizumab is not administered as an intravenous (IV) push or bolus injection. Infusions of study drug are prepared using aseptic technique. The patient's required dose of ravulizumab is further diluted into commercially available saline (0.9% sodium chloride; country-specific pharmacopeia) at the volume specified in Table 4. Ravulizumab admixture is administered to the patient using an IV tubing administration set via an infusion pump. Use of a 0.2 micron filter is required during infusion of ravulizumab.

TABLE 4

Dosing Reference Chart for Ravulizumab Dose Preparation							
Dose Type	Body Weight (kg) ^a	Dose (mg)	ravulizumab Volume (mL)	Saline Volume (mL)	Total Volume (mL)	Minimum Infusion Duration minutes (hours)	Maximum Infusion Rate (mL/hour)
Loading	≥5 to <10	600	60	60	120	228 (3.8)	31.5
	≥10 to <20	600	60	60	120	113 (1.9)	63.1
	≥20 to <30	900	90	90	180	86 (1.5)	120.0
	≥30 to <40	1200	120	120	240	77 (1.3)	184.6
	≥40 to <60	2400	240	240	480	114 (1.9)	253
	≥60 to <100	2700	270	270	540	102 (1.7)	318
	≥100	3000	300	300	600	108 (1.8)	333
Maintenance	≥5 to <10	300	30	30	60	113 (1.9)	31.5
	≥10 to <20	600	60	60	120	113 (1.9)	63.1
	≥20 to <30	2100	210	210	420	194 (3.3)	127.2
	≥30 to <40	2700	270	270	540	167 (2.8)	192.8
	≥40 to <60	3000	300	300	600	140 (2.4)	250
	≥60 to <100	3300	330	330	660	120 (2.0)	330
	≥100	3600	360	360	720	132 (2.2)	328

^aBody weight is obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.

supplied for clinical studies as a sterile, preservative-free 10 mg/mL solution in single-use vials and is designed for infusion by diluting into commercially available saline (0.9% sodium chloride injection; country-specific pharmacopeia) for administration via IV infusion (see Table 3).

TABLE 3

Study Drug	
Product Name	ravulizumab
Dosage Form	Concentrated solution (10 mg/mL) for infusion
Route of Administration	Intravenous infusion
Physical Description	Clear to translucent, slight whitish color, practically free from particles
Manufacturer	Alexion Pharmaceuticals, Inc. or Contracted Manufacturing Organization

Ravulizumab is packaged in US Pharmacopeia/European Pharmacopeia Type 1 borosilicate glass vials and stoppered with a butyl rubber stopper with an aluminum overseal and a flip-off cap. Study drug is supplied in kits.

Upon arrival of the study drug kits at the study site, the pharmacist or designee promptly removes the study drug kits from the shipping cooler and stores them in their original cartons under refrigerated conditions at 2 C to 8 C (35° F. to 47° F.) and protected from light. Ravulizumab is

Doses of study drug are only prepared and dispensed by qualified study personnel. Study drug is dispensed only to enrolled patients who are confirmed eligible for participation. Once study drug is prepared for a patient, it is only administered to that patient. Vials of study drug are for one-time use only and any drug product remaining in the vial is not used for another patient. Any drug remaining in the infusion tubing or infusion bag is not used for another patient.

^a Body weight is obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.

Patients receive a loading dose of ravulizumab on Day 1, followed by maintenance dosing of ravulizumab on Day 15 and once every eight weeks (q8w) thereafter for patients weighing ≥20 kg, or once every four weeks (q4w) for patients weighing <20 kg, as shown in Table 5. With the agreement of the Medical Monitor, the 600 mg loading dose can be given to patients weighing ≥5 to <10 kg as two separate infusions administered no more than 24 hours apart. In addition, administration of a supplemental dose of ravulizumab is permitted. Dosages are based on the patient's body weight recorded on the day of dosing or the most recently recorded weight. For patients entering the study on ecilizumab therapy, Day 1 of study treatment occurs 2 weeks from the patient's last dose of ecilizumab.

TABLE 5

Loading and Maintenance Treatment Regimens			
Body Weight Range (kg) ^a	Loading Dose (mg)	Maintenance Doses (mg)	Maintenance Dosing Frequency
≥5 to <10	600 ^b	300	q4w
≥10 to <20	600	600	q4w
≥20 to <30	900	2100	q8w
≥30 to <40	1200	2700	q8w
≥40 to <60	2400	3000	q8w
≥60 to <100	2700	3300	q8w
≥100	3000	3600	q8w

Abbreviations: q4w = once every 4 weeks; q8w = once every 8 weeks

^aDose regimen is based on body weight obtained at the study visit. If the study drug is prepared the night prior to the visit, the weight from the previous visit is used.

^bThe 600 mg loading dose can be given to patients weighing ≥5 to <10 kg as two separate infusions administered no more than 24 hours apart.

After the Primary Evaluation Period, all patients roll over into an Extension Period and continue their weight-based maintenance dose of ravulizumab on Day 183 and once every eight weeks (q8w) thereafter for patients weighing ≥20 kg, or once every four weeks (q4w) for patients weighing <20 kg, until the product is registered or approved (in accordance with country specific regulation) or for up to 2 years, whichever occurs first.

The actual time of all dose administrations, including any supplemental dose or dose administered as two separate infusions, is recorded. Due to its mechanism of action, the use of ravulizumab increases the patient's susceptibility to meningococcal infection (*N. meningitidis*). To reduce the risk of meningococcal infection, all patients are vaccinated against meningococcal infections within three years prior to, or at the time of, initiating study drug. Patients who initiate study drug treatment less than two weeks after receiving a meningococcal vaccine receive treatment with appropriate prophylactic antibiotics until two weeks after vaccination. Vaccines against serotypes A, C, Y, W135 and B, where

local practice on the appropriate use of antibacterial agents and vaccination. All patients are monitored for early signs of meningococcal infection, evaluated immediately if infection is suspected, and treated with appropriate antibiotics, if necessary. Patients are also vaccinated against Hib and *S. pneumoniae* according to national and local vaccination schedule guidelines.

7. Efficacy Assessments

Administration of a packed red blood cell (pRBC) transfusion, including the hemoglobin result and symptoms that triggered the transfusion, and the number of units transfused, are documented in the electronic case report form (eCRF).

Blood and urine samples are collected at the times indicated in the Schedule of Assessments.

The following disease-related laboratory parameters are measured during the study: LDH, free hemoglobin, occult blood, urine, total C5, haptoglobin, reticulocyte count, PNH RBC clone size evaluated by high-sensitivity flow cytometry, and estimated glomerular filtration rate (calculated using the Schwartz formula).

The Investigator or designee record for each patient the presence or absence of the following signs and symptoms of PNH: fatigue, chest pain, hemoglobinuria, abdominal pain, dyspnea, dysphagia, erectile dysfunction, and red/dark urine or hemoglobinuria.

The Pediatric FACIT-Fatigue scale is a 13-item questionnaire that assesses fatigue and its impact upon daily activities and function over the preceding 7 days. Each item is scored on a 5-point scale, and total scores range from 0 to 52, with higher score indicating better QoL. The questionnaire is self-reported by patients who were ≥8 years of age at the time of enrollment and reported by caregivers for patients who were ≥5 to <8 years of age at the time of enrollment. Patients <5 years of age are assessed. The Pediatric FACIT-Fatigue Scale is shown in Table 6.

TABLE 6

Pediatric FACIT-Fatigue Scale					
	None of the time	A little bit of the time	Some of the time	Most of the time	All of the time
I feel tired. . .	0	1	2	3	4
I have energy (or strength). . .	0	1	2	3	4
I could do my usual things at home. . .	0	1	2	3	4
I had trouble starting things because I was too tired. . .	0	1	2	3	4
I had trouble finishing things because I was too tired. . .	0	1	2	3	4
I needed to sleep during the day. . .	0	1	2	3	4
I got upset by being too tired to do the things I want to do. . .	0	1	2	3	4
Being tired made it hard for me to play or go out with my friends as much as I'd like. . .	0	1	2	3	4
I needed help doing my usual things at home. . .	0	1	2	3	4
I feel weak. . .	0	1	2	3	4
I was too tired to eat. . .	0	1	2	3	4
Being tired made me sad. . .	0	1	2	3	4
Being tired made me mad (angry). . .	0	1	2	3	4

available, are recommended to prevent common pathogenic meningococcal serotypes. Patients are vaccinated or revaccinated according to current national vaccination guidelines or local practice for vaccination use with eculizumab. Vaccination may not be sufficient to prevent meningococcal infection. Consideration is given per official guidance and

Major adverse vascular events (MAVEs) are assessed as part of the planned evaluation for adverse events (AEs). The description of the MAVE, including the method of diagnosis (e.g., magnetic resonance imaging, ultrasound, angiogram), date of diagnosis, and date resolved (or ongoing) is collected on the eCRF as part of the patient's medical history (prior to

baseline). A MAVE is defined as follows: thrombophlebitis/deep vein thrombosis, pulmonary embolus, myocardial infarction, transient ischemic attack, unstable angina, renal vein thrombosis, acute peripheral vascular occlusion, mesenteric/visceral vein thrombosis or infarction, mesenteric/visceral arterial thrombosis or infarction, hepatic/portal vein thrombosis (Budd-Chiari syndrome), cerebral arterial occlusion/cerebrovascular accident, cerebral venous occlusion, renal arterial thrombosis, gangrene (non-traumatic; non-diabetic), amputation (non-traumatic; non-diabetic), and dermal thrombosis.

8. Safety Assessments

A review of demographic parameters, including age, gender, race and ethnicity is performed. A complete medical history is taken and documented. Weight and height are recorded. Head circumference is recorded for patients who are ≤ 3 years of age. The patient's PNH medical history, including PNH symptoms, date of diagnosis, PNH clone size, pRBC transfusions and history of any MAVEs, are documented at the Screening visit.

The patient's medical history, including prior and concomitant conditions/disorders and transfusion history, is recorded at the Screening Visit. Medication (prescription or over the counter, including vitamins and/or herbal supplements) use within 28 days prior to the start of Screening is also be recorded. Details of prior treatment with eculizumab, including dose level and frequency, are collected. Meningococcal vaccination within 3 years prior to the first dose of study drug, and vaccination history for Hib and *S. pneumoniae* from birth, is also recorded.

A physical examination includes the following assessments: general appearance; skin; head, ear, eye, nose, and throat; neck; lymph nodes; chest; heart; abdominal cavity; limb; central nervous system; and musculoskeletal system. An abbreviated physical examination consists of a body system relevant examination based upon Investigator judgment and patient symptoms. Physical growth (height, weight and head circumference [the latter only in patients ≤ 3 years of age]) is assessed.

Vital sign measurements are taken after the patient has been resting for at least five minutes, and include systolic and diastolic blood pressure (BP; millimeters of mercury [mmHg]), heart rate (beats/minute), respiratory rate (breaths/minute), and temperature (degrees Celsius [C] or degrees Fahrenheit [° F]).

Samples for serum pregnancy, hematology, chemistry, coagulation, and urinalysis are performed at the times specified in the Schedule of Assessments. Specific laboratory assessments are provided in Table 7. Samples for laboratory assessments are collected before each study drug administration. An alternative blood sampling schedule for infants, for whom less blood volume is collected, must be used. If a suspected event of breakthrough hemolysis occurs, an unscheduled visit takes place at which a sample is collected for analysis of LDH and PK/PD by the central laboratory.

TABLE 7

Laboratory Assessments	
Hematology	Clinical Chemistry
Free hemoglobin	Alanine aminotransferase
Haptoglobin	Albumin
Hematocrit	Alkaline phosphatase
Hemoglobin	Aspartate aminotransferase
Mean corpuscular hemoglobin	Bicarbonate
Platelet count	Blood urea nitrogen

TABLE 7-continued

Laboratory Assessments	
RBC count	Calcium
RBC distribution width	Chloride
RBC mean corpuscular volume	Creatinine
Reticulocyte count	Gamma-glutamyltransferase
WBC count	Glucose
WBC differential	Lactate dehydrogenase
Coagulation Panel	Magnesium
International normalized ratio	Phosphorus
Partial thromboplastin time	Potassium
Prothrombin time	Sodium
Urinalysis	Total bilirubin (direct and indirect)
Appearance	Total protein
Bilirubin	Uric acid
Blood	Other
Color	Antidrug antibody
Glucose	Beta human chorionic gonadotropin
Ketone	(females of childbearing potential only)
Nitrite	Chicken RBC assay
pH	Free and total C5
Specific gravity	Pharmacokinetic assay
Urobilinogen	PNH clone size
Urine Chemistry	
Microalbumin	

Abbreviations: C5=complement component 5; PNH=paroxysmal nocturnal hemoglobinuria; RBC=red blood cell; WBC=white blood cell

It is anticipated that some laboratory values may be outside the normal value range due to the underlying disease. The Investigators should use their medical judgment when assessing the clinical significance of these values. Clinical significance is defined as any variation in laboratory measurements that has medical relevance and that results in a change in medical care. If clinically significant laboratory changes from baseline value are noted, the changes are documented as adverse events. The Investigator also assesses the relationship to study treatment for all clinically significant out-of-range values. The Investigator monitors the patient through additional laboratory assessments until (1) values have returned to the normal range or baseline level, or (2) in the judgment of the Investigator, values that are outside the normal range are not related to the administration of study drug or other protocol-specific procedures.

For females of childbearing potential (i.e., have achieved menarche), a serum or urine pregnancy test (beta-human chorionic gonadotropin [β -hCG]) is performed according to the Schedule of Assessments.

Blood samples are analyzed for the hematology parameters listed in Table 7. Blood samples are also analyzed for the serum chemistry parameters listed in Table 7. Indirect bilirubin is calculated from total and direct bilirubin values. Therefore, indirect bilirubin results are not available if direct bilirubin is below the limit of quantification.

Chemistry assessments are performed at the time points specified in the Schedule of Assessments. The estimated glomerular filtration rate is calculated using the Schwartz formula for all visits at which serum chemistries are collected. Blood samples are also analyzed for the coagulation parameters listed in Table 7.

Urine samples are analyzed for the parameters listed in Table 7. A microscopic examination of urine samples is performed if the results of the macroscopic analysis are abnormal.

For each patient, single 12-lead digital electrocardiograms (ECGs) are collected according to the Schedule of Assess-

ments. Patients are supine for approximately 5 to 10 minutes before ECG collection and remain supine (but awake) during ECG collection. The ECG is assessed to see whether it is within normal limits and to determine the clinical significance of the results.

Blood samples are collected to test for presence and titer of antidrug antibodies to ravulizumab in serum prior to study drug administration as indicated in the Schedule of Assessments. Further characterization of antibody responses is conducted as appropriate, including binding and neutralizing antibodies, PK/PD, safety and activity of ravulizumab.

An adverse event is any untoward medical occurrence in a patient administered a pharmaceutical product that does not necessarily have a causal relationship with this treatment. An adverse event can therefore, be any unfavorable or unintended sign (e.g., an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product.

Situations in which an untoward medical occurrence did not occur (e.g., hospitalization for elective surgery if planned before the start of the study, admissions for social reasons or for convenience), and anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen are not adverse events. Transfusions are treated as efficacy endpoints. Transfusions administered in the inpatient or outpatient setting should not be captured as adverse events or serious adverse events unless identified as such. Lack of drug effect is not an adverse event in clinical studies, because the purpose of the clinical study is to establish drug effect.

A medication error (including intentional misuse, abuse, and overdose of the product) or use other than what is defined in the protocol is not considered an adverse event unless there is an untoward medical occurrence as a result of a medication error.

Cases of pregnancy that occur during maternal or paternal exposure to investigational product are to be reported within 24 hours of Investigator/site awareness. Data on fetal outcome and breastfeeding is collected for regulatory reporting and safety evaluation.

The severity of adverse events is graded using Common Terminology Criteria for Adverse Events (CTCAE) version 4.03 or higher. A grading (severity) scale is provided for each adverse event term. Each CTCAE term is a Lowest Level Term (LLT) per the Medical Dictionary for Regulatory Activities (MedDRA®). Each LLT is coded to a MedDRA preferred term (PT). Grade refers to the severity of the adverse event. The CTCAE assigns a grade of 1 through 5, with unique clinical descriptions of severity for each adverse event, as set forth in Table 8.

TABLE 8

Adverse Event Severity Grading Scale	
Grade	Description
Grade 1	Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
Grade 2	Moderate; minimal, local or noninvasive intervention indicated; limiting age-appropriate instrumental ADL ^a
Grade 3	Severe or medically significant, but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL ^b
Grade 4	Life-threatening consequences; urgent intervention indicated.
Grade 5	Death related to AE.

Abbreviations: ADL = activities of daily living; AE = adverse event

^aInstrumental ADL refers to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

^bSelf-care ADL refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden.

Severity and seriousness are differentiated. Severity describes the intensity of an adverse event, while the term seriousness refers to an adverse event that has met specific criteria for a serious adverse event. A causality assessment (Unrelated, Unlikely, Possible, Probable, or Definite) is provided for all adverse events (both serious and nonserious) based upon the Investigator's medical judgment and the observed symptoms associated with the event, as set forth in Table 9.

TABLE 9

Causality Assessment Descriptions	
Assessment	Description
Not Related/ Unrelated	Suggests that there is no causal association between the investigational product and the reported event.
Unlikely Related	Suggests that the clinical picture is highly consistent with a cause other than the investigational product but attribution cannot be made with absolute certainty and a relationship between the investigational product and AE cannot be excluded with complete confidence.
Possibly Related	Suggests that treatment with the investigational product may have caused or contributed to the AE (i.e., the event follows a reasonable temporal sequence from the time of drug administration and/or follows a known response pattern to the investigational product, but could also have been produced by other factors).
Probably Related	Suggests that a reasonable temporal sequence of the event with the investigational product administration exists and the likely causal association of the event with the investigational product. This is based upon the known pharmacological action of the investigational product, known or previously reported adverse reactions to the investigational product or class of drugs, or judgment based on the Investigator's clinical experience.
Definitely Related	Temporal relationship to the investigational product, other conditions (concurrent illness, concurrent medication reaction, or progression/expression of disease state) do not appear to explain event, corresponds with the known pharmaceutical profile, improvement on discontinuation, reappearance on rechallenge.

A SAE is any untoward medical occurrence that: results in death, is life-threatening (i.e., patient was at risk of death at the time of the event), requires inpatient hospitalization or prolongation of existing hospitalization, results in persistent or significant disability/incapacity, or is a congenital anomaly/birth defect.

Important medical events that may not result in death, be immediately life-threatening, or require hospitalization are considered a serious adverse event when, based upon appropriate medical judgment, they jeopardize the patient or require intervention to prevent one of the outcomes listed above.

Suspected unexpected serious adverse reactions (SUSARs) are serious events that are not listed in the Investigator's Brochure and that the Investigator identifies as related to investigational product or procedure.

All adverse events are collected from the signing of the informed consent form (ICF) until 56 days after the last dose of study drug for patients with ET or until 56 days after the last dose of study drug for patients who complete the study. All adverse events are recorded on the eCRF upon becoming aware of their occurrence. Investigators are instructed to report the serious adverse event, including their assessment (e.g., severity, seriousness, and potential relatedness to study drug).

9. Pharmacokinetics (PK) and Pharmacodynamics (PD)

Blood samples for determination of serum drug concentrations and PD assessments are collected before and after administration of study drug at the time points indicated in the Schedule of Assessments. The actual date and time (24-hour clock time) of each sampling is recorded. In the event of breakthrough hemolysis, an additional PK/PD sample is required.

End of infusion blood samples for PK and PD assessment is collected from the arm opposite to the arm used for infusing drug. Assessments for PK/PD are as follows: serum ravulizumab concentration over time, free and total C5 concentrations over time, and cRBC hemolytic activity over time

10. Statistical Methods and Planned Analyses

A clinical study report (CSR) is produced based on efficacy, safety, PK, PD and immunogenicity data collected through the end of the 26-week Primary Evaluation Period (Day 183). A final CSR to summarize long-term efficacy, safety, PK, PD and immunogenicity parameters is produced at study completion.

This study is descriptive in nature and not statistically powered for hypothesis testing due to the rarity of disease in pediatric patients. Approximately 12 patients are enrolled to ensure at least 10 evaluable patients complete the 26-week period. A sample size of 10 is expected to be sufficient to adequately describe PK/PD in pediatric patients with PNH. Efficacy analyses are performed on the Full Analysis Set (FAS). The FAS includes all patients who received at least one dose of ravulizumab and have at least one post-baseline assessment.

Safety analyses are performed on the Safety Set, defined as all patients who receive at least one dose of ravulizumab.

Pharmacokinetic and PD analyses are performed on all patients who receive at least one dose of ravulizumab and who have evaluable PK and PD data.

Patient demographic and baseline characteristics, including medical history and transfusion history, are summarized for the clinical study report (FAS) and Safety sets. All patients are included in the summaries of patient disposition, which describe the frequency and percentage of patients

screened and treated and who completed or withdraw from the study, along with reason for withdrawal from the study.

The numbers of patients who are treated, discontinue treatment (along with reason for treatment discontinuation), complete or withdraw from the Primary Evaluation Period (along with reason for withdrawal), enter the Extension Period, and complete or withdraw from the Extension Period (along with reason for withdrawal) are tabulated.

Each patient's prior and concomitant medication use is coded using the World Health Organization Drug Dictionary, and the frequency and percentage of concomitant medications is summarized. Medications are summarized by Anatomic-Therapeutic-Chemical class and generic name using frequency counts and percentages of patients in the Safety set. The number of infusions received per patient is tabulated for the FAS and Safety sets. Efficacy analyses is performed using the FAS. Continuous variables are summarized using descriptive statistics, including number of observations and mean, SD, median, minimum and maximum values. Categorical variables are summarized by frequency counts and percentage of patients. Analyses are conducted separately for naïve and previously ecilizumab-treated patients.

The following definitions are used for adverse events. A pretreatment adverse event is any adverse event that starts after providing informed consent, but before the first infusion of study drug. A treatment-emergent adverse event is any adverse event that starts during or after the first infusion of study drug. A treatment-emergent serious adverse event is a treatment-emergent AE (TEAE) that is serious.

The incidence of TEAEs, TEAEs leading to withdrawal from the study, TEAEs leading to study treatment discontinuation, drug-related TEAEs, TEAEs during study drug administration, severe TEAEs and SAEs are summarized. All adverse events are coded using MedDRA version 18 or higher, and summarized by system organ class and preferred term. Detailed by-patient listings of TEAEs, SAEs, related TEAEs, TEAEs during study drug administration, TEAEs leading to withdrawal from the study and TEAEs leading to study treatment discontinuation are provided.

Adverse changes from baseline in physical examination findings are classified as adverse events and analyzed accordingly.

Vital signs are summarized descriptively at baseline and postbaseline time points and for changes from baseline by treatment group.

Height, weight and head circumference (the latter only for patients <3 years of age) are summarized descriptively at baseline and post-baseline time points and for changes from baseline by treatment group.

Changes in clinical chemistry, hematology, and urinalysis results from baseline to post-baseline study time points are summarized descriptively. Shift tables over time are presented for all central laboratory values, where applicable, using normal, low or high based on normal range values. Listings of patients with abnormal results are provided. By-patient data listings of ECG parameters are provided. Changes from baseline in electrocardiogram intervals (PR, RR, QT and QTcF) are provided. QT interval is corrected for heart rate using Fridericia's formula (QTcF).

Incidence and titers for antidrug antibodies (ADAs) to ravulizumab are summarized in tabular format. Individual serum concentration data for all patients who receive at least one dose of ravulizumab and who have evaluable PK data are used to derive PK parameters for ravulizumab.

Graphs of mean serum concentration-time profiles are constructed. Graphs of serum concentration-time profiles for

individual patients are also provided. Actual dose administration and sampling times is used for all calculations. Descriptive statistics are calculated for serum concentration data at each sampling time, as appropriate. Assessment of population-PK is considered using data from this study or in combination with data from other studies.

PD analyses are performed for all patients who receive at least one dose of ravulizumab and who have evaluable PD data.

Descriptive statistics are presented for all ravulizumab PD endpoints at each sampling time. The PD effects of ravuli-

zumab administered IV are evaluated by assessing the absolute values and changes and percentage changes from baseline in total and free C5 serum concentrations and change from baseline in cRBC hemolysis over time, as appropriate.

Assessments of ravulizumab PK/PD relationships are explored using data from this study or in combination with data from other studies. Analyses are conducted separately for naïve and previously eculizumab-treated patients.

If a Day 1 assessment is missing, the Screening assessment is used as the baseline assessment.

SEQUENCE SUMMARY	
GYIFSNYWIQ	SEQ ID NO: 1
EILPGSGSTEYTENFKD	SEQ ID NO: 2
YFFGSSPNWYFDV	SEQ ID NO: 3
GASENIYGALN	SEQ ID NO: 4
GATNLAD	SEQ ID NO: 5
QNVLNTPLT	SEQ ID NO: 6
QVQLVQSGAE VKKPGASVKV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE	SEQ ID NO: 7
ILPGSGSTEY TENFKDRVTM TRDISTSTVY MELSSLRSED TAVYYCARYF	
FGSSPNWYFD VWGQGTLLTVV SS	
DIQMTQSPSS LSASVGDRVT ITCGASENIY GALNWKQKP GKAPKLLIYG	SEQ ID NO: 8
ATNLADGVPS RFSGSGSGTD FTLTISSLQP EDFATYYCQN VLNTPLTFGQ	
GTKVEIK	
ASTKGPSVFPP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALISGV	SEQ ID NO: 9
HTFPAVLQSS GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS NTKVDKTVET	
KCCVECPPEP APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVSQEDP	
EVQFNWYVDG VEVHNAKTKP REEQFNSTYR VVSVLIVLHQ DWINGKEYKC	
KVSNKGLPSS IEKTISKAKG QPREPQVYTL PPSQEEMTKN QVSLTCLVKG	
FYPSTDAVEW ESNGQPENNY KTTTPVLDSD GSFFLYSRLT VDKSRWQEGN	
VFSCSVMEHA LHNHYTQKSL SLSLGK	
QVQLVQSGAE VKKPGASVKV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE	SEQ ID NO: 10
ILPGSGSTEY TENFKDRVTM TRDISTSTVY MELSSLRSED TAVYYCARYF	
FGSSPNWYFD VWGQGTLLTVV SSASTKGPSV FPLAPCSRST SSTAALGCL	
VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV VTPSSNFGT	
QTYTCNVDHK PSNTKVDKTV ERKCCVECPP CPAPPVAGPS VFLFPPKPKD	
TLMISRTPEV TCVVDVSQEDPEVQFNWYV DGVEVHNAKT KPREEQFNST	
YRVVSVLTIVL HQDWINGKEY KCKVSNKGLP SSIEKTISKA KGQPREPQVY	

SEQUENCE SUMMARY

TLPPSQEEMT NQVSLTCLVK GFYPSDIAVE WESNGQPENN YKITPPVLDS
 DGSFFLYSRL TVDKSRWQEG NVFSCSVMHE ALHNHYTQKS LSLSLGK
 SEQ ID NO: 11
 DIQMTQSPSS LSASVGDRVIT ITCGASENIY GALNWYQQKP GKAPKLLIYG
 ATNLADGVPS RFGSGSGSTD FTLTISSLQP EDFATYYCQN VLNTPLTFGQ
 GTKVEIKRTV AAPSVFIFPP SDEQLKSGTA SVVCLLNNFY PREAKVQWKV
 DNALQSGNSQ ESVTEQDSKD STYLSSTLT LSKADYEKKH VYACEVTHQG
 LSSPVTKSEN RGEC
 SEQ ID NO: 12
 QVQLVQSGAE VKKPGASVKV SCKASGHIFS NYWIQWVRQA PGQGLEWMGE
 ILPGSGHTEY TENFKDRVIM TRDTSTSTVY MELSSLRSED TAVYYCARYF
 FGSSPNWYFD VWGQGTLLTV SS
 SEQ ID NO: 13
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV
 HTFPAVLQSS GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS NTKVDKTVR
 KCCVECPPCP APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVSQEDP
 EVQFNWYVDG VEVHNAKTKP REEQFNSTYR VVSVLIVLHQ DWLNGKEYKC
 KVSNGKLPSS IEKTISKAKG QPREPQVYTL PPSQEEMTKN QVSLTCLVKG
 FYPSDIAVEW ESNGQPENNY KTTTPVLDSG GSFFLYSRLT VDKSRWQEGN
 VFSCSVLHEA LHSHTQKSL SLSLGK
 SEQ ID NO: 14
 QVQLVQSGAE VKKPGASVKV SCKASGHIFS NYWIQWVRQA PGQGLEWMGE
 ILPGSGHTEY TENFKDRVIM TRDISTSTVY MELSSLRSED TAVYYCARYF
 FGSSPNWYFD VWGQGTLLTV SSASTKGPSV FPLAPCSRST SESTAALGCL
 VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV VTPSSNFGT
 QTYTCNVDHK PSNTKVDKTV ERKCCVECPP CPAPPVAGPS VFLFPPKPKD
 TLMISRTPEV TCVVVDVSQE DPEVQFNWYV DGVEVHNAKT KPREEQFNST
 YRVVSVLIVL HQDWLNGKEY KCKVSNKGLP SSIEKTISKA KGQPREPQVY
 TLPPSQEEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPVLDS
 SDGSFFLYSR LTVDKSRWQE GNVFSCSVLH EALHSHYTQK SLSLSLGK
 SEQ ID NO: 15
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV
 HTFPAVLQSS GLYSLSSVVT VISSNFGTQT YTCNVDHKPS NTKVDKTVR
 KCCVECPPCP APPVAGPSVF LFPPKPKDTL YITREPEVTC VVVDVSHEDP
 EVQFNWYVDG MEVHNAKTKP REEQFNSTER VVSVLTVVHQ DWLNGKEYKC
 KVSNGKLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN QVSLTCLVKG
 FYPSDIAVEW ESNGQPENNY KTTTPMLDSG GSFFLYSKLT VDKSRWQQGN
 VFSCSVMHEA LHNHYTQKSL SLSPGK
 SEQ ID NO: 16
 QVQLVQSGAE VKKPGASVKV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE
 ILPGSGSTAY TENFKDRVIM TRDISTSTVY MELSSLRSED TAVYYCARYF

-continued

SEQUENCE SUMMARY

FGSSPNWYFD VWGQGLTVTV SSASTKGPSV FPLAPCSRST SESTAALGCL	
VKDYPPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV VTVTSSNFGT	
QTYTCNVDPK PSNTKVDKTV ERKCCVECPP CPAPPVAGPS VFLPPPKPKD	
TLYITREPEV TCVVVDVSHE DPEVQFNWYV DGMEVHNAKT KPREEQENST	
FRVSVLIVV HQDWLNGKEY KCKVSNKGLP APIEKTISK KGQPREPQVY	
TLPPSREEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPMLD	
SDGSFFLYSK LTVDKSRWQQ GNVFSCSVMH EALHNHYTQK SLSLSPGK	
	SEQ ID NO: 17
GASENIYHALN	
	SEQ ID NO: 18
EILPGSGHTEYTENFKD	
	SEQ ID NO: 19
GHIFSNIYWIQ	
	SEQ ID NO: 20
QVQLVQSGAE VKKPGASVKV SCKASGHIFS NYWIQWVRQA PGQGLEWMGE	
ILPGSGHTEY TENFKDRVTM TRDTSTSTVY MELSSLRSED TAVYYCARYF	
FGSSPNWYFD VWGQGLTVTV SSASTKGPSV FPLAPCSRST SESTAALGCL	
VKDYPPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV VTPSSNFGT	
QTYTCNVDPK PSNTKVDKTV ERKCCVECPP CPAPPVAGPS VFLPPPKPKD	
TLMISRTPEV TCVVVDVSQE DPEVQFNWYV DGVEVHNAKT KPREEQFNST	
YRVSVLTVL HQDWINGKEY KCKVSNKGLP SSIKTISKA KGQPREPQVY	
TLPPSQEEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPVLD	
SDGSFFLYSR LTVDKSRWQE GNVFSCSVMH EALHNHYTQK SLSLSLGK	
	SEQ ID NO: 21
SYAIS	
	SEQ ID NO: 22
GIGPFFGTANYAQKFQG	
	SEQ ID NO: 23
DTPYFDY	
	SEQ ID NO: 24
SGDSIPNYYVY	
	SEQ ID NO: 25
DDSNRPS	
	SEQ ID NO: 26
QSFDSLNAEV	
	SEQ ID NO: 27
QVQLVQSGAE VKKPGSSVKV SCKASGGTFS SYAISVWRQA PGQGLEWMGG	
IGPFFGTANY AQKFQGRVTI TADESTSTAY MELSSLRSED TAVYYCARDT	
PYFDYWGGGT LVTVSS	
	SEQ ID NO: 28
DIELTQPPSV SVAPGQTARI SCSGDSIPNY YVYVYQKPG QAPVLVIYDD	
SNRPSGIPER FSGSNSGNTA TLTISGTQAE DEADYYCQSF DSSLNAEVEG	
GGTKLTVL	
	SEQ ID NO: 29
NYIS	

SEQUENCE SUMMARY	
IIDPDDSYTEYSP SFQG	SEQ ID NO: 30
YEYGGFDI	SEQ ID NO: 31
SGDNIGNSYVH	SEQ ID NO: 32
KDNRPS	SEQ ID NO: 33
GTYDIESYV	SEQ ID NO: 34
EVQLVQSGAE VKKPGESLKI SCKGSGYSFT NYISWVRQMP GKGLEWMGII	SEQ ID NO: 35
DPDDSYTEYS PSFQGQVTIS ADKSISTAYL QWSSLKASDT AMYYCAREY	
GGFDIWGGT LVTVSS	
SYELTQPPSV SVAPGQTARI SCSGDNIGNS YVHWYQQKPG QAPVLVIYKD	SEQ ID NO: 36
NDRPSGIPER FSGSNSGNTA TLTISGTQAE DEADYYCGTY DIESYVFGGG	
TKLTVL	
SSYYVA	SEQ ID NO: 37
AIYTGSGATYKASWAKG	SEQ ID NO: 38
DGGYDYPHAMHY	SEQ ID NO: 39
QASQNIGSSLA	SEQ ID NO: 40
GASKTHS	SEQ ID NO: 41
QSTKVGSSYGNH	SEQ ID NO: 42
QVQLVESGGG LVQPGGSLRL SCAASGETSH SSYYVAWVRQ APGKLEWVG	SEQ ID NO: 43
AIYTGSGATY KASWAKGRFT ISKDISKNQV VLTMINMDPV DTATYYCASD	
GGYDYPHAM HYWGQGTLLT VSS	
DVMTQSPSS LSASVGDRVT ITCQASQNIG SSLAWYQQK QAPRLLIYG	SEQ ID NO: 44
ASKTHSGVPS RFSGSGSGTD FTLTISSLQP EDVATYYCQS TKVGSSYGNH	
FGGGTKVEIK	
QVQLVESGGG LVQPGSLRL SCAASGFTVH SSYYMAWVRQ APGKLEWVG	SEQ ID NO: 45
AIFTGSGAEY KAEWAKGRVT ISKDISKNQV VLTMINMDPV DTATYYCASD	
AGYDYPHAM HYWGQGTLLT VSSASTKGPS VFPLAPSSKS TSGGTAALGC	
LVKDYFPEPV TVSWNSGALT SGVHTFPAVL QSSGLYSLSS VVTVPSSSLG	
TQTYICNVNH KPSNTKVDKK VEPKSCDKTH TCPPCPAPEL RRGPKVELFP	
PKPKDTLMIS RTPEVTCVVV DVSHEDPEVK FNWYVDGVEV HNAKIKPREE	
QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKGLPSSIEK TISKAKGQPR	
EPQVYTLPPS REEMTKNQVS LTCLVKGFYP SDIAVEWESN GPENNYKTT	

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SEQUENCE SUMMARY

PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSVLHEALHA HYTRKELSLS

P

SEQ ID NO: 46

DIQMTQSPSS LSASVGDVRT ITCRASQGIS SSLAWYQQKP GKAPKLLIYG

ASETESGVPS RFGSGSGTD FTLTISSLQP EDFATYYCQN TKVGSSYGNT

FGGGTKVEIK RTVAAPSVFI FPPSDEQLKS GTASVVCLLN NFYPREAKVQ

WKVDNALQSG NSQESVTEQD SKDSTYSLSS TLILSKADYE KHKVYACEVT

HQGLSSPVTK SFNRGEC

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 46

<210> SEQ ID NO 1

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 1

Gly Tyr Ile Phe Ser Asn Tyr Trp Ile Gln

1 5 10

<210> SEQ ID NO 2

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 2

Glu Ile Leu Pro Gly Ser Gly Ser Thr Glu Tyr Thr Glu Asn Phe Lys

1 5 10 15

Asp

<210> SEQ ID NO 3

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 3

Tyr Phe Phe Gly Ser Ser Pro Asn Trp Tyr Phe Asp Val

1 5 10

<210> SEQ ID NO 4

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 4

Gly Ala Ser Glu Asn Ile Tyr Gly Ala Leu Asn
1 5 10

<210> SEQ ID NO 5

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 5

Gly Ala Thr Asn Leu Ala Asp
1 5

<210> SEQ ID NO 6

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 6

Gln Asn Val Leu Asn Thr Pro Leu Thr
1 5

<210> SEQ ID NO 7

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic polypeptide"

<400> SEQUENCE: 7

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Ser Asn Tyr
20 25 30

Trp Ile Gln Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35 40 45

Gly Glu Ile Leu Pro Gly Ser Gly Ser Thr Glu Tyr Thr Glu Asn Phe
50 55 60

Lys Asp Arg Val Thr Met Thr Arg Asp Thr Ser Thr Ser Thr Val Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Tyr Phe Phe Gly Ser Ser Pro Asn Trp Tyr Phe Asp Val Trp
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 8

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic polypeptide"

```

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<400> SEQUENCE: 8

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```

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15

```

```

Asp Arg Val Thr Ile Thr Cys Gly Ala Ser Glu Asn Ile Tyr Gly Ala
                20           25           30

```

```

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
                35           40           45

```

```

Tyr Gly Ala Thr Asn Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
50           55           60

```

```

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65           70           75           80

```

```

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Asn Val Leu Asn Thr Pro Leu
                85           90           95

```

```

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100           105

```

```

<210> SEQ ID NO 9

```

```

<211> LENGTH: 326

```

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<212> TYPE: PRT

```

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<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

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<221> NAME/KEY: source

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```

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic polypeptide"

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<400> SEQUENCE: 9

```

```

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1           5           10           15

```

```

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20           25           30

```

```

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35           40           45

```

```

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50           55           60

```

```

Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
65           70           75           80

```

```

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
85           90           95

```

```

Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro
100           105           110

```

```

Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
115           120           125

```

```

Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
130           135           140

```

```

Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly
145           150           155           160

```

```

Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn
165           170           175

```

```

Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp
180           185           190

```

```

Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro
195           200           205

```

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Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu
 210 215 220
 Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn
 225 230 235 240
 Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
 245 250 255
 Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
 260 265 270
 Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg
 275 280 285
 Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys
 290 295 300
 Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
 305 310 315 320
 Ser Leu Ser Leu Gly Lys
 325

<210> SEQ ID NO 10
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 10

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Ser Asn Tyr
 20 25 30
 Trp Ile Gln Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45
 Gly Glu Ile Leu Pro Gly Ser Gly Ser Thr Glu Tyr Thr Glu Asn Phe
 50 55 60
 Lys Asp Arg Val Thr Met Thr Arg Asp Thr Ser Thr Ser Thr Val Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Tyr Phe Phe Gly Ser Ser Pro Asn Trp Tyr Phe Asp Val Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
 115 120 125
 Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr
 130 135 140
 Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
 145 150 155 160
 Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
 165 170 175
 Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
 180 185 190
 Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp
 195 200 205
 His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys
 210 215 220
 Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser

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225	230	235	240
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg	245	250	255
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro	260	265	270
Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala	275	280	285
Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val	290	295	300
Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr	305	310	315
Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr	325	330	335
Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu	340	345	350
Pro Pro Ser Gln Glu Glu Met Thr Asn Gln Val Ser Leu Thr Cys Leu	355	360	365
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn	370	375	380
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser	385	390	395
Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg	405	410	415
Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu	420	425	430
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys	435	440	445

<210> SEQ ID NO 11
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 11

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly	1	5	10	15
Asp Arg Val Thr Ile Thr Cys Gly Ala Ser Glu Asn Ile Tyr Gly Ala	20	25	30	
Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile	35	40	45	
Tyr Gly Ala Thr Asn Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly	50	55	60	
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro	65	70	75	80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Asn Val Leu Asn Thr Pro Leu	85	90	95	
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala	100	105	110	
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly	115	120	125	
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala	130	135	140	

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Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
 145 150 155 160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
 165 170 175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
 180 185 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
 195 200 205

Phe Asn Arg Gly Glu Cys
 210

<210> SEQ ID NO 12
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 12

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly His Ile Phe Ser Asn Tyr
 20 25 30

Trp Ile Gln Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Glu Ile Leu Pro Gly Ser Gly His Thr Glu Tyr Thr Glu Asn Phe
 50 55 60

Lys Asp Arg Val Thr Met Thr Arg Asp Thr Ser Thr Ser Thr Val Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Tyr Phe Phe Gly Ser Ser Pro Asn Trp Tyr Phe Asp Val Trp
 100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 13
 <211> LENGTH: 326
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 13

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
 1 5 10 15

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
 65 70 75 80

-continued

100					105					110					
Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro
	115						120					125			
Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr
	130					135					140				
Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr
	145					150					155				160
Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro
			165						170					175	
Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr
		180						185					190		
Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp
		195					200					205			
His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys
	210					215					220				
Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser
	225					230					235				240
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
			245						250					255	
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro
		260						265					270		
Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
		275					280					285			
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val
	290					295					300				
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
	305					310					315				320
Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr
			325						330					335	
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
			340					345					350		
Pro	Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
		355					360					365			
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
	370					375					380				
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
	385					390					395				400
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser
			405						410					415	
Arg	Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Leu	His	Glu	Ala
		420						425					430		
Leu	His	Ser	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys
		435					440					445			

<210> SEQ ID NO 15

<211> LENGTH: 326

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic polypeptide"

<400> SEQUENCE: 15

Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg
1				5								10			15

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Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30
 Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45
 Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60
 Leu Ser Ser Val Val Thr Val Thr Ser Ser Asn Phe Gly Thr Gln Thr
 65 70 75 80
 Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85 90 95
 Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro
 100 105 110
 Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
 115 120 125
 Thr Leu Tyr Ile Thr Arg Glu Pro Glu Val Thr Cys Val Val Val Asp
 130 135 140
 Val Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly
 145 150 155 160
 Met Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn
 165 170 175
 Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp
 180 185 190
 Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro
 195 200 205
 Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu
 210 215 220
 Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
 225 230 235 240
 Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
 245 250 255
 Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
 260 265 270
 Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
 275 280 285
 Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
 290 295 300
 Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
 305 310 315 320
 Ser Leu Ser Pro Gly Lys
 325

<210> SEQ ID NO 16
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 16

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Ser Asn Tyr
 20 25 30

Trp	Ile	Gln	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met
354045															
Gly	Glu	Ile	Leu	Pro	Gly	Ser	Gly	Ser	Thr	Glu	Tyr	Thr	Glu	Asn	Phe
505560															
Lys	Asp	Arg	Val	Thr	Met	Thr	Arg	Asp	Thr	Ser	Thr	Ser	Thr	Val	Tyr
65707580															
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
859095															
Ala	Arg	Tyr	Phe	Phe	Gly	Ser	Ser	Pro	Asn	Trp	Tyr	Phe	Asp	Val	Trp
100105110															
Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro
115120125															
Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr
130135140															
Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr
145150155160															
Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro
165170175															
Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr
180185190															
Val	Thr	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp
195200205															
His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys
210215220															
Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser
225230235240															
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Tyr	Ile	Thr	Arg
245250255															
Glu	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
260265270															
Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Met	Glu	Val	His	Asn	Ala
275280285															
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val
290295300															
Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
305310315320															
Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr
325330335															
Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
340345350															
Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
355360365															
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
370375380															
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp
385390395400															
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser
405410415															
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
420425430435															
Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys
435440445															

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```

<210> SEQ ID NO 17
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 17

```

```

Gly Ala Ser Glu Asn Ile Tyr His Ala Leu Asn
1           5           10

```

```

<210> SEQ ID NO 18
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 18

```

```

Glu Ile Leu Pro Gly Ser Gly His Thr Glu Tyr Thr Glu Asn Phe Lys
1           5           10           15

```

```

Asp

```

```

<210> SEQ ID NO 19
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 19

```

```

Gly His Ile Phe Ser Asn Tyr Trp Ile Gln
1           5           10

```

```

<210> SEQ ID NO 20
<211> LENGTH: 448
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic polypeptide"

```

```

<400> SEQUENCE: 20

```

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10           15

```

```

Ser Val Lys Val Ser Cys Lys Ala Ser Gly His Ile Phe Ser Asn Tyr
20           25           30

```

```

Trp Ile Gln Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35           40           45

```

```

Gly Glu Ile Leu Pro Gly Ser Gly His Thr Glu Tyr Thr Glu Asn Phe
50           55           60

```

```

Lys Asp Arg Val Thr Met Thr Arg Asp Thr Ser Thr Ser Thr Val Tyr
65           70           75           80

```

```

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85           90           95

```

```

Ala Arg Tyr Phe Phe Gly Ser Ser Pro Asn Trp Tyr Phe Asp Val Trp
100          105          110

```

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Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro
	115						120					125			
Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr
	130					135					140				
Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr
	145				150					155					160
Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro
			165						170					175	
Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr
		180						185					190		
Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp
	195						200					205			
His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys
	210					215					220				
Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser
	225				230					235					240
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
			245						250					255	
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro
		260						265					270		
Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
	275						280					285			
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val
	290					295					300				
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
	305				310					315					320
Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr
			325						330					335	
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
		340						345					350		
Pro	Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
		355					360					365			
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
	370					375					380				
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
	385				390					395					400
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser
			405						410					415	
Arg	Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
		420						425					430		
Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys
	435						440					445			

<210> SEQ ID NO 21
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic peptide"

<400> SEQUENCE: 21

Ser Tyr Ala Ile Ser
 1 5

-continued

```

<210> SEQ ID NO 22
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 22

```

```

Gly Ile Gly Pro Phe Phe Gly Thr Ala Asn Tyr Ala Gln Lys Phe Gln
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 23
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 23

```

```

Asp Thr Pro Tyr Phe Asp Tyr
1           5

```

```

<210> SEQ ID NO 24
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 24

```

```

Ser Gly Asp Ser Ile Pro Asn Tyr Tyr Val Tyr
1           5           10

```

```

<210> SEQ ID NO 25
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 25

```

```

Asp Asp Ser Asn Arg Pro Ser
1           5

```

```

<210> SEQ ID NO 26
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 26

```

```

Gln Ser Phe Asp Ser Ser Leu Asn Ala Glu Val
1           5           10

```

```

<210> SEQ ID NO 27
<211> LENGTH: 116

```

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic polypeptide"

```

```

<400> SEQUENCE: 27

```

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1             5             10             15

```

```

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Ser Ser Tyr
      20             25             30

```

```

Ala Ile Ser Val Trp Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
      35             40             45

```

```

Gly Gly Ile Gly Pro Phe Phe Gly Thr Ala Asn Tyr Ala Gln Lys Phe
      50             55             60

```

```

Gln Gly Arg Val Thr Ile Thr Ala Asp Glu Ser Thr Ser Thr Ala Tyr
      65             70             75             80

```

```

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
      85             90             95

```

```

Ala Arg Asp Thr Pro Tyr Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val
      100            105            110

```

```

Thr Val Ser Ser
      115

```

```

<210> SEQ ID NO 28
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic polypeptide"

```

```

<400> SEQUENCE: 28

```

```

Asp Ile Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Gln
 1             5             10             15

```

```

Thr Ala Arg Ile Ser Cys Ser Gly Asp Ser Ile Pro Asn Tyr Tyr Val
      20             25             30

```

```

Tyr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
      35             40             45

```

```

Asp Asp Ser Asn Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
      50             55             60

```

```

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Glu
      65             70             75             80

```

```

Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Phe Asp Ser Ser Leu Asn Ala
      85             90             95

```

```

Glu Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
      100            105

```

```

<210> SEQ ID NO 29
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic peptide"

```

```

<400> SEQUENCE: 29

```

```

Asn Tyr Ile Ser

```

-continued

1

```

<210> SEQ ID NO 30
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 30

```

```

Ile Ile Asp Pro Asp Asp Ser Tyr Thr Glu Tyr Ser Pro Ser Phe Gln
1           5              10              15

```

```

Gly

```

```

<210> SEQ ID NO 31
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 31

```

```

Tyr Glu Tyr Gly Gly Phe Asp Ile
1           5

```

```

<210> SEQ ID NO 32
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 32

```

```

Ser Gly Asp Asn Ile Gly Asn Ser Tyr Val His
1           5              10

```

```

<210> SEQ ID NO 33
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 33

```

```

Lys Asp Asn Asp Arg Pro Ser
1           5

```

```

<210> SEQ ID NO 34
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic peptide"

```

```

<400> SEQUENCE: 34

```

```

Gly Thr Tyr Asp Ile Glu Ser Tyr Val
1           5

```

-continued

```

<210> SEQ ID NO 35
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic polypeptide"

```

```

<400> SEQUENCE: 35

```

```

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1          5          10          15
Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Asn Tyr
20          25          30
Ile Ser Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met Gly
35          40          45
Ile Ile Asp Pro Asp Asp Ser Tyr Thr Glu Tyr Ser Pro Ser Phe Gln
50          55          60
Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr Leu
65          70          75          80
Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys Ala
85          90          95
Arg Tyr Glu Tyr Gly Gly Phe Asp Ile Trp Gly Gln Gly Thr Leu Val
100         105         110
Thr Val Ser Ser
115

```

```

<210> SEQ ID NO 36
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic polypeptide"

```

```

<400> SEQUENCE: 36

```

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Gln
1          5          10          15
Thr Ala Arg Ile Ser Cys Ser Gly Asp Asn Ile Gly Asn Ser Tyr Val
20          25          30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
35          40          45
Lys Asp Asn Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50          55          60
Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Glu
65          70          75          80
Asp Glu Ala Asp Tyr Tyr Cys Gly Thr Tyr Asp Ile Glu Ser Tyr Val
85          90          95
Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100         105

```

```

<210> SEQ ID NO 37
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic peptide"

```

-continued

<400> SEQUENCE: 37

Ser Ser Tyr Tyr Val Ala
1 5

<210> SEQ ID NO 38
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 38

Ala Ile Tyr Thr Gly Ser Gly Ala Thr Tyr Lys Ala Ser Trp Ala Lys
1 5 10 15

Gly

<210> SEQ ID NO 39
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 39

Asp Gly Gly Tyr Asp Tyr Pro Thr His Ala Met His Tyr
1 5 10

<210> SEQ ID NO 40
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 40

Gln Ala Ser Gln Asn Ile Gly Ser Ser Leu Ala
1 5 10

<210> SEQ ID NO 41
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 41

Gly Ala Ser Lys Thr His Ser
1 5

<210> SEQ ID NO 42
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic peptide"

<400> SEQUENCE: 42

-continued

Gln Ser Thr Lys Val Gly Ser Ser Tyr Gly Asn His
1 5 10

<210> SEQ ID NO 43
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 43

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Ser His Ser Ser
20 25 30
 Tyr Tyr Val Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
35 40 45
 Val Gly Ala Ile Tyr Thr Gly Ser Gly Ala Thr Tyr Lys Ala Ser Trp
50 55 60
 Ala Lys Gly Arg Phe Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val
65 70 75 80
 Val Leu Thr Met Thr Asn Met Asp Pro Val Asp Thr Ala Thr Tyr Tyr
85 90 95
 Cys Ala Ser Asp Gly Gly Tyr Asp Tyr Pro Thr His Ala Met His Tyr
100 105 110
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 44
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic polypeptide"

<400> SEQUENCE: 44

Asp Val Val Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asn Ile Gly Ser Ser
20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45
 Tyr Gly Ala Ser Lys Thr His Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
 Glu Asp Val Ala Thr Tyr Tyr Cys Gln Ser Thr Lys Val Gly Ser Ser
85 90 95
 Tyr Gly Asn His Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 45
 <211> LENGTH: 451
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic polypeptide"

<400> SEQUENCE: 45

Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Arg	1	5	10	15
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Val	His	Ser	Ser	20	25	30	
Tyr	Tyr	Met	Ala	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	35	40	45	
Val	Gly	Ala	Ile	Phe	Thr	Gly	Ser	Gly	Ala	Glu	Tyr	Lys	Ala	Glu	Trp	50	55	60	
Ala	Lys	Gly	Arg	Val	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	65	70	75	80
Val	Leu	Thr	Met	Thr	Asn	Met	Asp	Pro	Val	Asp	Thr	Ala	Thr	Tyr	Tyr	85	90	95	
Cys	Ala	Ser	Asp	Ala	Gly	Tyr	Asp	Tyr	Pro	Thr	His	Ala	Met	His	Tyr	100	105	110	
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	115	120	125	
Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	130	135	140	
Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	145	150	155	160
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	165	170	175	
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	180	185	190	
Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	195	200	205	
Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	210	215	220	
Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	225	230	235	240
Arg	Arg	Gly	Pro	Lys	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	245	250	255	
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	260	265	270	
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	275	280	285	
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	290	295	300	
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	305	310	315	320
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	325	330	335	
Ser	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	340	345	350	
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	355	360	365	
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	370	375	380	
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	385	390	395	400

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```

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
    405                                410                415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
    420                                425                430

Val Leu His Glu Ala Leu His Ala His Tyr Thr Arg Lys Glu Leu Ser
    435                                440                445

Leu Ser Pro
    450

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
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Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ser
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Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35         40         45

Tyr Gly Ala Ser Glu Thr Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50         55         60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65         70         75         80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Asn Thr Lys Val Gly Ser Ser
85         90         95

Tyr Gly Asn Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr
100        105        110

Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu
115        120        125

Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro
130        135        140

Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly
145        150        155        160

Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr
165        170        175

Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His
180        185        190

Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val
195        200        205

Thr Lys Ser Phe Asn Arg Gly Glu Cys
210        215

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What is claimed is:

1. A method of treating a human pediatric patient with Paroxysmal Nocturnal Hemoglobinuria (PNH), the method comprising administering to the patient an effective amount of an anti-C5 antibody comprising heavy and light chains comprising the amino acid sequences shown in SEQ ID NOs:14 and 11, respectively, wherein the anti-C5 antibody is administered intravenously:

(a) once on Day 1 at a dose of 600 mg to a patient weighing ≥ 5 to <10 kg, 600 mg to a patient weighing

≥ 10 to <20 kg, 900 mg to a patient weighing ≥ 20 to <30 kg, or 1200 mg to a patient weighing ≥ 30 to <40 kg; and

(b) on Day 15 and every four weeks thereafter at a dose of 300 mg to a patient weighing ≥ 5 to <10 kg or 600 mg to a patient weighing ≥ 10 to <20 kg; or on Day 15 and every eight weeks thereafter at a dose of 2100 mg to a patient weighing ≥ 20 to <30 kg, or 2700 mg to a patient weighing ≥ 30 to <40 kg.

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2. The method of claim 1, wherein the anti-C5 antibody binds to human C5 at pH 7.4 and 25 C with an affinity dissociation constant (K_D) that is in the range $0.1 \text{ nM} \leq K_D \leq 1 \text{ nM}$.

3. The method of claim 1, wherein the anti-C5 antibody binds to human C5 at pH 6.0 and 25 C with a $K_D \geq 10 \text{ nM}$. 5

4. The method of claim 1, wherein the anti-C5 antibody is administered to a patient weighing ≥ 5 to $<10 \text{ kg}$:

(a) once on Day 1 at a dose of 600 mg; and

(b) on Day 15 and every four weeks thereafter at a dose of 300 mg. 10

5. The method of claim 1, wherein the anti-C5 antibody is administered to a patient weighing ≥ 10 to $<20 \text{ kg}$:

(a) once on Day 1 at a dose of 600 mg; and

(b) on Day 15 and every four weeks thereafter at a dose of 600 mg. 15

6. The method of claim 1, wherein the anti-C5 antibody is administered to a patient weighing ≥ 20 to $<30 \text{ kg}$:

(a) once on Day 1 at a dose of 900 mg; and

(b) on Day 15 and every eight weeks thereafter at a dose of 2100 mg. 20

7. The method of claim 1, wherein the anti-C5 antibody is administered to a patient weighing ≥ 30 to $<40 \text{ kg}$:

(a) once on Day 1 at a dose of 1200 mg; and

(b) on Day 15 and every eight weeks thereafter at a dose of 2700 mg. 25

8. The method of claim 1, further comprising measuring a serum trough concentration of the anti-C5 antibody to maintain a serum trough concentration of the anti-C5 antibody of $100 \text{ } \mu\text{g/mL}$ or greater in the patient. 30

9. The method of claim 1, wherein the method comprises an administration cycle comprising a total of 26 weeks of treatment.

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10. The method of claim 1, further comprising determining that the treated patient has:

(a) terminal complement inhibition;

(b) a reduction of hemolysis as assessed by lactate dehydrogenase (LDH) levels compared to baseline;

(c) produces at least one therapeutic effect selected from the group consisting of: a reduction or cessation in abdominal pain, dyspnea, dysphagia, chest pain and erectile dysfunction compared to baseline;

(d) a shift toward normal levels of at least one hemolysis related hematologic biomarker selected from the group consisting of: free hemoglobin, haptoglobin, reticulocyte count, PNH red blood cell (RBC) clone and D dimer;

(e) a reduction in the need for blood transfusions compared to baseline;

(f) a reduction in major adverse vascular events (MAVEs);

(g) a shift toward normal levels of estimated glomerular filtration rate (eGFR) or spot urine:albumin:creatinine and plasma brain natriuretic peptide (BNP); and/or

(h) a change from baseline in quality of life, assessed via the Functional Assessment of Chronic Illness Therapy (FACIT) Fatigue Scale, version 4 and the European Organisation for Research and Treatment of Cancer, Quality of Life Questionnaire Core 30 Scale compared to baseline.

11. The method of claim 1, wherein the patient has previously been treated with eculizumab and Day 1 of the treatment is two weeks or more from the patient's last dose of eculizumab.

12. The method of claim 1, wherein the anti-C5 antibody is ravulizumab.

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